

ZTE ZXA10 C300 MANUAL



Table of Contents

[Table of Contents](#)

[Preface](#)

[Introduction](#)

[Overview](#)

[Alarm Level](#)

[Alarm List](#)

[Alarm Configuration](#)

[Setting Alarm Preferences](#)

[Figure 1 Alarm Sound Setup](#)

[Figure 2 Alarm Color Setup](#)

Customizing Alarm Levels

Figure 3 Forward Template Setup

Figure 4 New Severity

Setting Alarm Maintenance Suggestions

Figure 5 Modify Status

Figure 6 Maintenance Suggestion Setting

Figure 7 Function Buttons

Setting Alarm Rules

Figure 8 Management Rule Setting

Setting the Alarm Threshold

Figure 9 Perf Threshold Admin

Figure 10 Add Performance Threshold

Figure 11 Perf threshold admin

Real-Time Monitoring

Real-Time Alarm Monitoring

Real-Time Event Monitoring

Figure 12 Real-Time Alarms

Real-Time Notification Monitoring

Figure 13 Real-Time Events

Alarm Information Query

Querying Current Alarms

Figure 14 Real-Time Notifications

Figure 15 Current Alarm Query Conditions

Querying History Alarms

Figure 16 Query Current Alarm

Figure 17 History Alarm Query Conditions

Figure 18 Query History Alarm

Collecting History Alarm Statistics

Figure 19 History Alarm Statistics - Report Management

Querying Events

Figure 20 History Alarm Statistics - List of Report Templates

Querying Notifications

Figure 21 Query Event

Figure 22 Query Notification

EPON Service Alarms

Device OLT PortBer State

PON LOS

ONU Errored Symbol Period Event

ONU Errored Frame Period Event

ONU Errored Frame Period Event

ONU Errored Frame Seconds Summary Event

ONU Uplink Bit Error

ONU Uplink Frame Error

ONU Optical Power Threshold Crossing

ONU Offline

ONU Dying Gasp

OLT Reset

Local ONU Configure Failed (OLT)

Extended OAM

Extended OAM Discovery Failed (ONU)

Unknown ONU Online

Unknown ONU Offline (ONU)

Deny Unknown ONU Register (ONU)

ONU Request Auto Config

CES Service Alarms

PDH Interface LOS Alarm

PDH Interface AIS Alarm

PW Local Alarm

PW Remote Alarm

Optical Interface LOS Alarm

Loss of Frame Alarm

Out of Frame Alarm

MS-AIS

MS-RDI

RS SF

RS SD

AU-LOP

AU AIS

HP TIM

HP UNEQ

HPRDI

HP PLM

HP-AIS

HP TU LOM

TU-LOP

TU LOM

TU-AIS

LP TIM

LP UNEQ

LPRDI

LP PLM

System Alarms

Cold Startup Notification

Active/Standby Card Switchover Notification

Backup Data Synchronization Failure Notification

CPU Overload Alarm

Memory Overload Alarm

[Card Status Error Alarm](#)

[Terminal Login Notification](#)

[Terminal Logout Notification](#)

[NM Heartbeat Notification](#)

[MAC Table Utilization Exceeding Threshold Notification](#)

[Card Version Update Failure Alarm](#)

[Uplink Port Down Alarm](#)

[Access Port Up Notification](#)

[Access Port Down Notification](#)

[Uplink Port Output Bandwidth Occupancy Exceeding Threshold Alarm](#)

[Uplink Port Input Bandwidth Occupancy Exceeding Threshold Alarm](#)

[Threshold Alarm](#)

[High Temperature Alarm](#)

[Fan Card Link Down Alarm](#)

[Fan Fault Alarm](#)

[Over-voltage Power Alarm](#)

[Under-voltage Power Alarm](#)

[NO Input Power Alarm](#)

[High Temperature Card Alarm](#)

[CCM Notification](#)

[Figures](#)

[Tables](#)

[Index](#)

[Glossary](#)

[Other Manuals](#)[Lib Projects](#)



ZXA10 C300

EPON Optical Access Convergence Equipment

Alarms Manual

Version 1.1

ZTE CORPORATION
NO. 55, Hi-tech Road South, ShenZhen, P.R.China
Postcode: 518057
Tel: (86) 755 26771900
Fax: (86) 755 26770801
URL: <http://ensupport.zte.com.cn>
E-mail: support@zte.com.cn

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Contents

Preface	I
Introduction	1
Overview	1
Alarm Level	1
Alarm List	1
Alarm Configuration	4
Setting Alarm Preferences	4
Customizing Alarm Levels	7
Setting Alarm Maintenance Suggestions	9
Setting Alarm Rules	11
Setting the Alarm Threshold	12
Real-Time Monitoring	14
Real-Time Alarm Monitoring	14
Real-Time Event Monitoring	15
Real-Time Notification Monitoring	16
Alarm Information Query	17
Querying Current Alarms	17
Querying History Alarms	19
Collecting History Alarm Statistics	21
Querying Events	22
Querying Notifications	23
EPON Service Alarms	25
Device OLT PortBer State	25
PON LOS	26
ONU Errored Symbol Period Event	26
ONU Errored Frame Period Event	27
ONU Errored Frame Period Event	28
ONU Errored Frame Seconds Summary Event	28
ONU Uplink Bit Error	29
ONU Uplink Frame Error	29
ONU Optical Power Threshold Crossing	30

ONU Offline	30
ONU Dying Gasp	31
OLT Reset.....	31
Local ONU Configure Failed (OLT)	32
Extended OAM	32
Extended OAM Discovery Failed (ONU)	35
Unknown ONU Online.....	36
Unknown ONU Offline (ONU)	36
Deny Unknown ONU Register (ONU).....	36
ONU Request Auto Config	37
CES Service Alarms	39
PDH Interface LOS Alarm.....	39
PDH Interface AIS Alarm	40
PW Local Alarm	41
PW Remote Alarm.....	41
Optical Interface LOS Alarm	42
Loss of Frame Alarm	42
Out of Frame Alarm	43
MS-AIS	44
MS-RDI.....	44
RS SF	45
RS SD.....	45
AU-LOP	46
AU AIS.....	46
HP TIM.....	47
HP UNEQ.....	47
HPRDI.....	48
HP PLM	49
HP-AIS.....	49
HP TU LOM	50
TU-LOP	50
TU LOM	51
TU-AIS.....	51
LP TIM	52
LP UNEQ	52
LPRDI	53
LP PLM	54
System Alarms	55
Cold Startup Notification.....	55
Active/Standby Card Switchover Notification	56

Backup Data Synchronization Failure Notification	56
CPU Overload Alarm.....	57
Memory Overload Alarm	57
Card Status Error Alarm	58
Terminal Login Notification.....	59
Terminal Logout Notification	59
NM Heartbeat Notification	59
MAC Table Utilization Exceeding Threshold Notification	60
Card Version Update Failure Alarm	61
Uplink Port Down Alarm.....	61
Access Port Up Notification.....	62
Access Port Down Notification.....	62
Uplink Port Output Bandwidth Occupancy Exceeding Threshold Alarm	63
Uplink Port Input Bandwidth Occupancy Exceeding Threshold Alarm	64
High Temperature Alarm.....	64
Fan Card Link Down Alarm.....	65
Fan Fault Alarm	65
Over-voltage Power Alarm	66
Under-voltage Power Alarm.....	66
NO Input Power Alarm.....	67
High Temperature Card Alarm	67
CCM Notification	68
Figures	69
Tables	71
Index	73
Glossary	75

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Preface

Purpose	This manual provides description about the alarms related to the ZXA10 C300 EPON Optical Access Convergence Equipment.										
Intended Audience	This document is intended for engineers and technicians who perform operation and maintenance activities of the ZXA10 C300 EPON Optical Access Convergence Equipment.										
Prerequisite Skill and Knowledge	To use this document effectively, users should have a general understanding of network technology. Familiarity with the following is helpful: ■ The ZXA10 C300 system and its various components ■ Maintenance procedures ■ Local operating procedures										
What Is in This Manual	This manual contains the following chapters: <table><tr><th>Chapter</th><th>Summary</th></tr><tr><td>Chapter 1, Introduction</td><td>Describes the alarm levels, the alarm list, the alarm settings, the real-time monitoring and the alarm information check.</td></tr><tr><td>Chapter 2, EPON Service Alarms</td><td>Describes the EPON service alarms.</td></tr><tr><td>Chapter 3, CES Service Alarms</td><td>Describes the CES service alarms.</td></tr><tr><td>Chapter 4, System Alarms</td><td>Describes the system alarms.</td></tr></table>	Chapter	Summary	Chapter 1, Introduction	Describes the alarm levels, the alarm list, the alarm settings, the real-time monitoring and the alarm information check.	Chapter 2, EPON Service Alarms	Describes the EPON service alarms.	Chapter 3, CES Service Alarms	Describes the CES service alarms.	Chapter 4, System Alarms	Describes the system alarms.
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Chapter 2, EPON Service Alarms	Describes the EPON service alarms.										
Chapter 3, CES Service Alarms	Describes the CES service alarms.										
Chapter 4, System Alarms	Describes the system alarms.										
Related Documentation	The following documentation is related to this manual: ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Documentation Guide</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Feature Description</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Product Information</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Hardware Description</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Hardware Installation (19" Shelf)</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Hardware Installation (21" Shelf)</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Configuration Manual (CLI)</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Configuration Manual (NetNumen)</i> ■ <i>ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Routine Maintenance</i>										

- *ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Command Manual (Volume I)*
- *ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Command Manual (Volume II)*
- *ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Command Manual (Volume III)*
- *ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Command Manual (Volume IV)*
- *ZXA10 C300 (V1.1) EPON Optical Access Convergence Equipment Command Manual (Volume V)*

Conventions ZTE documents employ the following typographical conventions.

TABLE 1 TYPOGRAPHICAL CONVENTIONS

Typeface	Meaning
<i>Italics</i>	References to other Manuals and documents.
"Quotes"	Links on screens.
Bold	Menus, menu options, function names, input fields, radio button names, check boxes, drop-down lists, dialog box names, window names.
CAPS	Keys on the keyboard and buttons on screens and company name.
Constant width	Text that you type, program code, files and directory names, and function names.
[]	Optional parameters
{ }	Mandatory parameters
I	Select one of the parameters that are delimited by it.
 Note:	Note: Provides additional information about a certain topic.
 Checkpoint:	Checkpoint: Indicates that a particular step needs to be checked before proceeding further.
 Tip:	Tip: Indicates a suggestion or hint to make things easier or more productive for the reader.

Mouse Operation Conventions are as follows:

TABLE 2 MOUSE OPERATION CONVENTIONS

Typeface	Meaning
Click	Refers to clicking the primary mouse button (usually the left mouse button) once.
Double-click	Refers to quickly clicking the primary mouse button (usually the left mouse button) twice.
Right-click	Refers to clicking the secondary mouse button (usually the right mouse button) once.
Drag	Refers to pressing and holding a mouse button and moving the mouse.

How to Get in Touch

The following sections provide information on how to obtain support for the documentation and the software.

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Chapter 1

Introduction

Table of Contents

Overview.....	1
Alarm Configuration	4
Real-Time Monitoring	14
Alarm Information Query	17

Overview

Alarms are the prompt information raised when exceptions or faults occur during the device operation. Generally, an alarm lasts for a certain period until the exception or fault is removed.

This manual describes the alarms that are displayed and queried in the ZTE NetNumen N31 [NMS](#).

Alarm Level

According to alarm severities, alarms can be grouped into five levels:

- Critical alarm
- Major alarm
- Minor alarm
- Warning alarm
- Notification

The first four levels can be configured as required. For the configuration details, refer to the section 'Customizing Alarm Levels'.

Notifications are defined by the system and cannot be modified. If the user wants to modify a notification to a higher alarm level, contact the ZTE technical support engineers.

Alarm List

[Table 3](#) provides the ZXA10 C300 alarm list.

TABLE 3 ZXA10 C300 ALARM LIST

Alarm Code	Alarm Information	Alarm Level
EPON Service Alarm		
43014	Device OLT port BER state	Critical
43015	PON LOS	Critical
43017	ONU errored symbol period event	Critical
43018	ONU errored frame event	Critical
43019	ONU errored frame period event	Critical
43020	ONU errored frame seconds summary event	Critical
43021	ONU uplink bit error	Critical
43022	ONU uplink frame error	Critical
43009	ONU optical power threshold crossing	Critical
43016	ONU offline	Critical
43061	ONU dying gasp	Critical
43049	OLT reset	Notification
43053	Local ONU configure failed	Notification
43060	Extended OAM	Notification
43052	Extended OAM discovery failed	Notification
43058	Unknown ONU online	Notification
43054	Unknown ONU offline	Notification
43055	Deny unknown ONU register	Notification
43062	ONU request auto config	Notification
CES Service Alarm		
36865	TDM interface LOS	Critical
36866	TDM interface AIS	Critical
36867	PW local alarm	Major
36868	PW remote alarm	Critical
36869	LOS	Critical
36870	LOF	Critical
36871	OOF	Critical

Alarm Code	Alarm Information	Alarm Level
36872	MS AIS	Critical
36873	MS RDI	Critical
36874	RS SF	Warning
36875	RS SD	Warning
36876	AU LOP	Critical
36877	AU AIS	Critical
36878	HPTIM	Warning
36879	HP UNEQ	Critical
36880	HP RDI	Critical
36881	HP PLM	Warning
36882	HP AIS	Critical
36883	HP TU LOM	Critical
36884	TU LOP	Critical
36885	TU LOM	Critical
36886	TU AIS	Critical
36887	LP TIM	Warning
36888	LP Uneq	Critical
36889	LP RDI	Critical
36890	LP PLM	Warning
System Alarm		
33025	Device coldstart	Notification
33026	Master card swapped	Notification
33027	Backup data synchronization failed Notification	Notification
33035	CPU overload alarm	Minor
33036	Memory overload alarm	Minor
33037	Card status error alarm	Critical
34816	Terminal log in notification	Notification
34817	Terminal log out notification	Notification
34818	NM heartbeat notification	Notification

Alarm Code	Alarm Information	Alarm Level
35584	MAC table usage over thresh trap	Notification
38401	Card version update failed alarm	Warning
40713	Uplink port down alarm	Critical
40714	Access port up notification	Notification
40715	Access port down notification	Notification
40718	Ether interface outbandwidth threshold alarm	Minor
40719	Ether interface inbandwidth threshold alarm	Minor
50944	High temperature alarm	Warning
50945	EPM monitor interface broken	Warning
50948	Fan fault alarm	Warning
50952	Power overvoltage alarm	Warning
50953	Power low voltage alarm	Warning
50954	No power input alarm	Warning
50955	Card high temperature alarm	Warning
55552	Ethernet connectivity detect alarm	Notification

Alarm Configuration

The ZTE NetNumen N31 [NMS](#) supports the customized configuration on the generation, levels, rules, handling, and threshold of the alarm. It includes the configuration on the alarm sound, the alarm color, the presetting message forwarding and the mails format.

This topic includes the following:

- Setting Alarm Preferences
- Customizing Alarm Levels
- Setting Alarm Maintenance Suggestions
- Setting Alarm Rules
- Setting the Alarm Threshold

Setting Alarm Preferences

Short Description Perform this procedure to set alarm preferences.

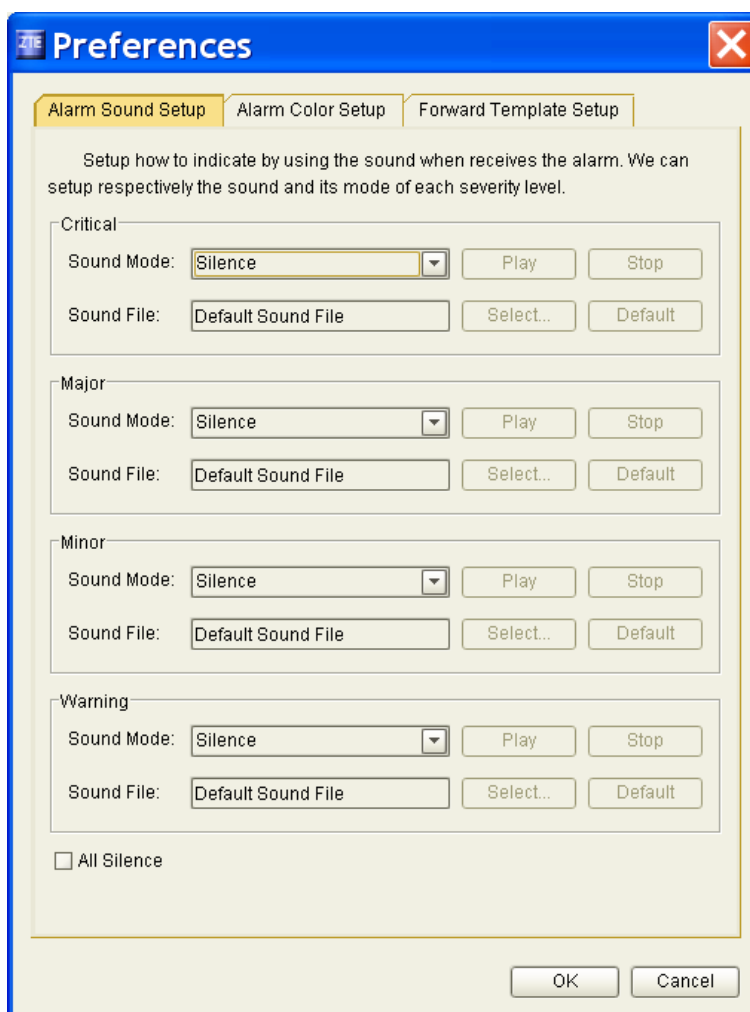
Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

- Context**
- Alarm sound setting enables the system to generate a customized sound and remind the user of faults and events in a user preferred mode.
 - Alarm color settings enables the system to generate a customized color to remind the user of faults and events levels in a user preferred mode.
 - Forwarding message and mail text style facilitates user to forward the alarms to the specified mailbox or handset through mails or messages.

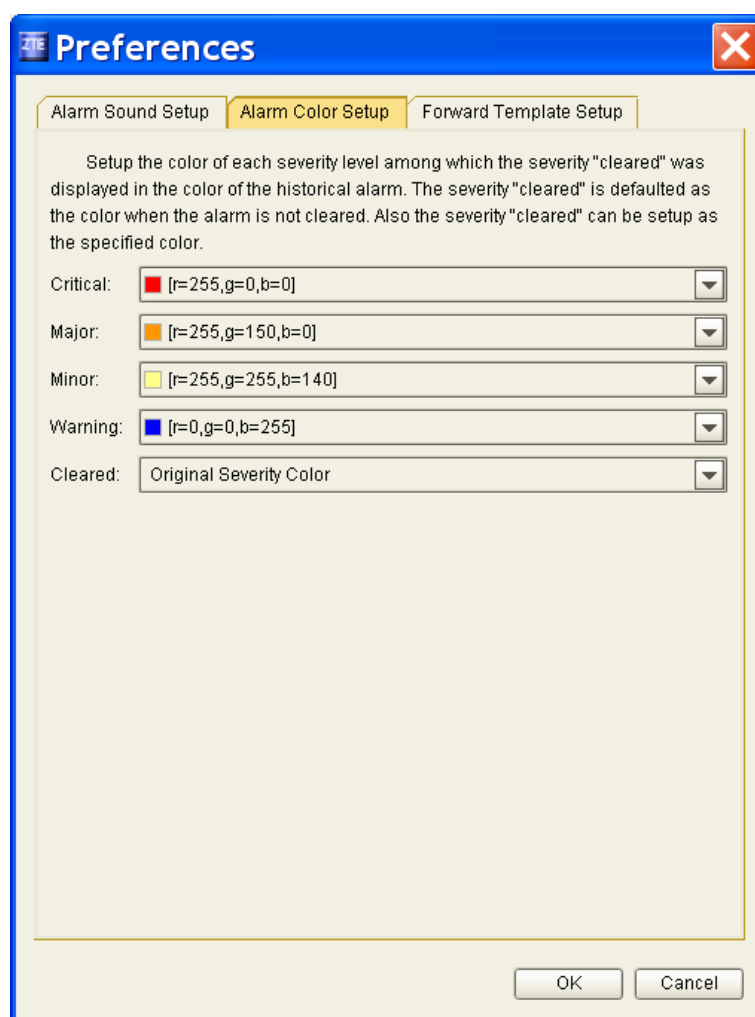
To set alarm preferences, perform the following steps:

- Steps**
1. Choose **View > Fault Management** to open the **Fault Management** interface.
 2. In the **Fault Management** interface, choose **Setting > Preferences**. The **Preferences** interface appears, as shown in [Figure 1](#).

FIGURE 1 ALARM SOUND SETUP

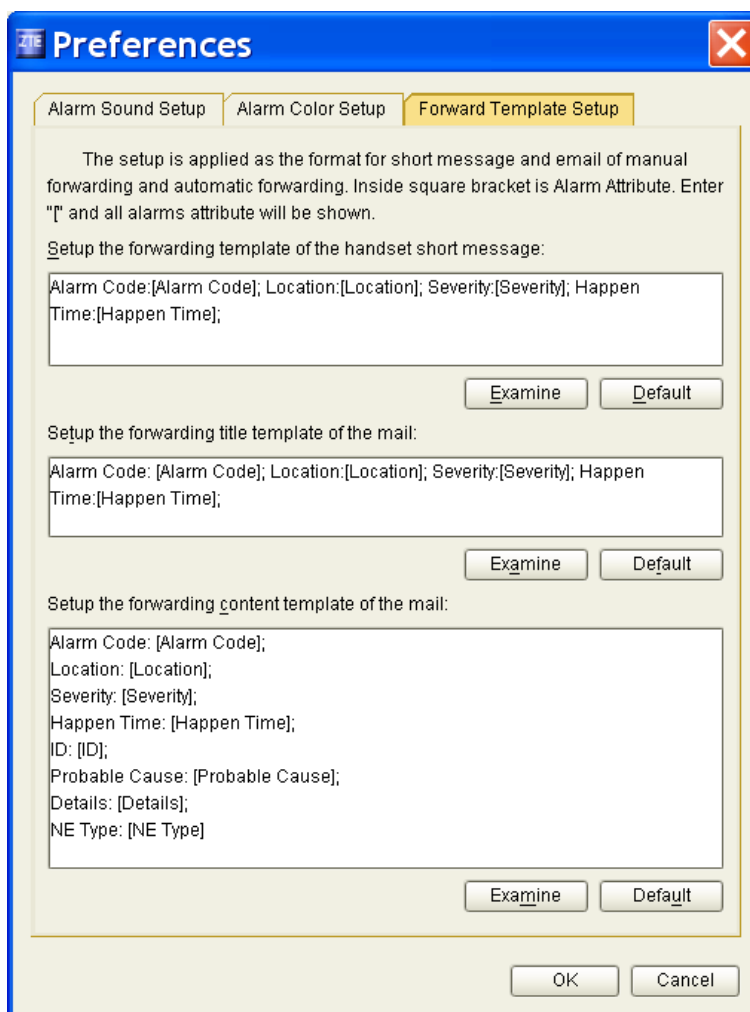


3. Click the **Alarm Sound Setup** tab to set the sound mode and the sound file, and play the selected sound. The sound mode includes:
 - ▶ Silence
 - ▶ Continuous Sound
 - ▶ Short Beeps
4. Click the **Alarm Color Setup** tab to set the color for each alarm level, as shown in [Figure 2](#).

FIGURE 2 ALARM COLOR SETUP

5. Click the **Forward Template Setup** tab to set the forwarding message and mail text style, as shown in [Figure 3](#).

FIGURE 3 FORWARD TEMPLATE SETUP



6. Click **Examine** to customize the forwarding mail or message template. Click **Default** to restore the template to the default.
7. Click **OK** to save the settings.

END OF STEPS

Result The alarm preferences are set.

Customizing Alarm Levels

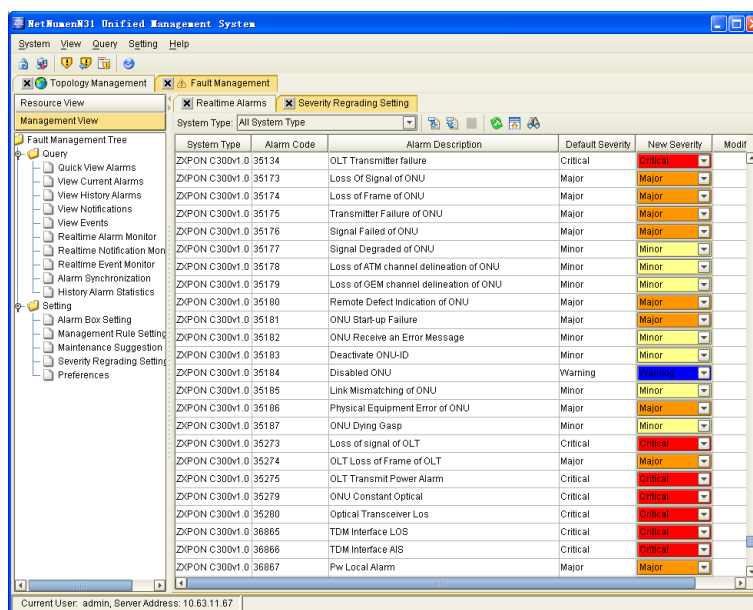
Short Description Perform this procedure to customize the alarm levels.

Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context Modify the alarm levels corresponding to the alarm codes to modify the alarm levels of the ZXA10 C300.

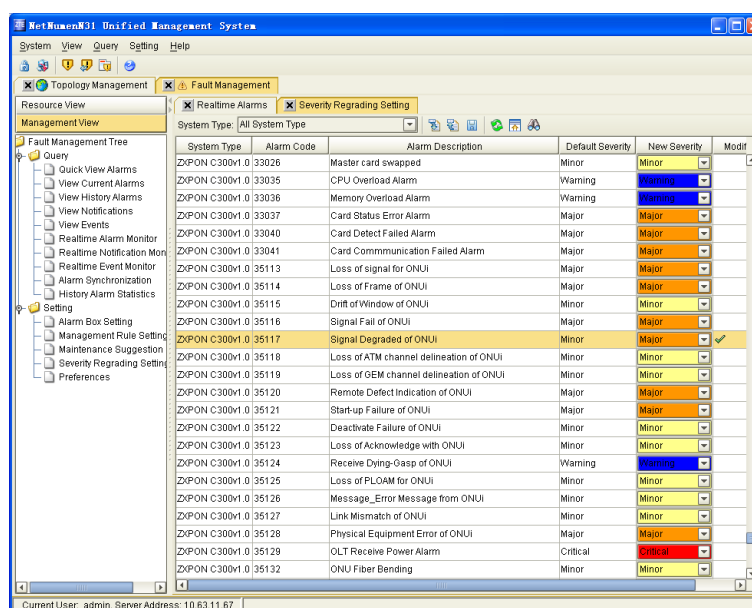
To customize the alarm levels, perform the following steps:

- Steps**
1. Choose **View > Fault Management** to open the **Fault Management** interface.
 2. In the **Fault Management** interface, choose **Setting > Severity Regrading Setting** to open the **Severity Regrading Setting** interface.
 3. Select **System Type**. All the alarm codes of this system type are displayed.
 4. To modify the level of an alarm, select the alarm. In the **New Severity** column, select a new severity, as shown in [Figure 4](#).

FIGURE 4 NEW SEVERITY

5. In the **Modify Status** column, a check appears, as shown in [Figure 5](#).

FIGURE 5 MODIFY STATUS



The alarm level on the user interface has been redefined. However, the information on the server has not been modified. To upload this modification, click  on the tool bar.

END OF STEPS

Result The alarm levels are customized.

Setting Alarm Maintenance Suggestions

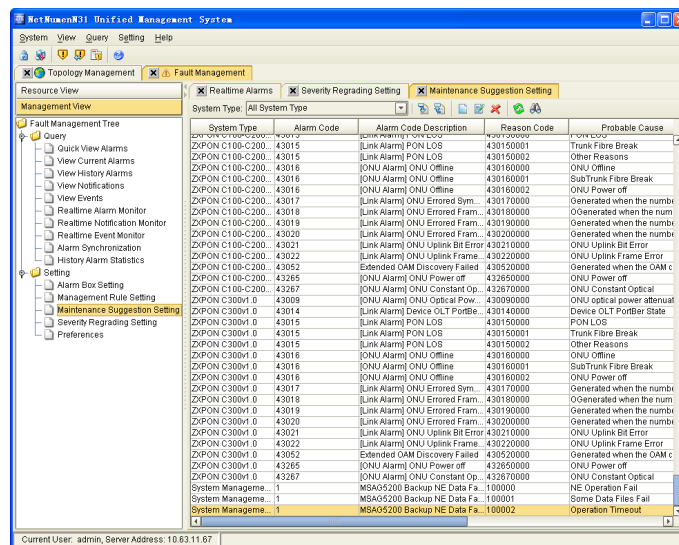
Short Description Perform this procedure to set alarm maintenance suggestions.

Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context To set alarm maintenance suggestions, perform the following steps:

- Steps**
1. Choose **View > Fault Management** to open the **Fault Management** interface.
 2. In the **Fault Management** interface, choose **Setting > Maintenance Suggestion Setting** to open the **Maintenance Suggestion Setting** interface, as shown in [Figure 6](#).

FIGURE 6 MAINTENANCE SUGGESTION SETTING



- The alarm maintenance suggestion list is displayed according to the system type. To view the alarm maintenance suggestions of another system type, select the system type from the **System Type** drop-down list.
- All the NMS alarm maintenance suggestions based on ZTE UNM can be imported into one another as long as the alarm code definition consistent are consistent.
- The function buttons appear in the maintenance suggestion view, as shown in [Figure 7](#).

FIGURE 7 FUNCTION BUTTONS



- The buttons include (from the left to the right) Import, Export, New, Modify, Delete, Refresh, and Find. [Table 4](#) provides the button descriptions.

TABLE 4 BUTTON DESCRIPTION

Button	Description
Import	Import the alarm maintenance suggestions into this system from the local computer.
Export	Export the alarm maintenance suggestions from this system to the local computer.
New	Create detailed information on the alarm maintenance suggestion.
Modify	Modify the information on a selected alarm maintenance suggestion.

Button	Description
Delete	Delete the detailed information on a selected alarm maintenance suggestion.
Refresh	Refresh the alarm maintenance suggestion information.
Find	Find the alarm maintenance suggestion information.

END OF STEPS

Result The alarm maintenance suggestions are set.

Setting Alarm Rules

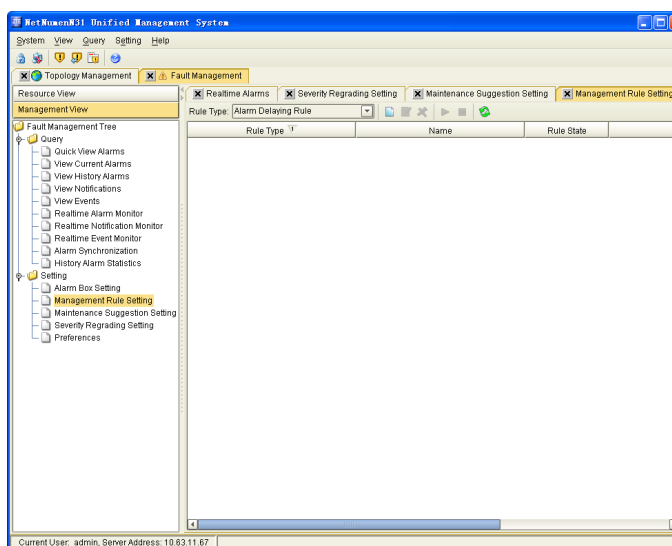
Short Description Perform this procedure to set alarm rules.

Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZX10 C300 properly.

Context To set alarm rules, perform the following steps:

- Steps**
1. Choose **View > Fault Management** to open the **Fault Management** interface.
 2. In the **Fault Management** interface, choose **Setting > Management Rule Setting** to open the **Management Rule Setting** interface, as shown in [Figure 8](#).

FIGURE 8 MANAGEMENT RULE SETTING



3. In the **Management Rule Setting** interface, select a rule type. Right-click the rule type and select the operation.

END OF STEPS

Result The alarm rules are set.

Setting the Alarm Threshold

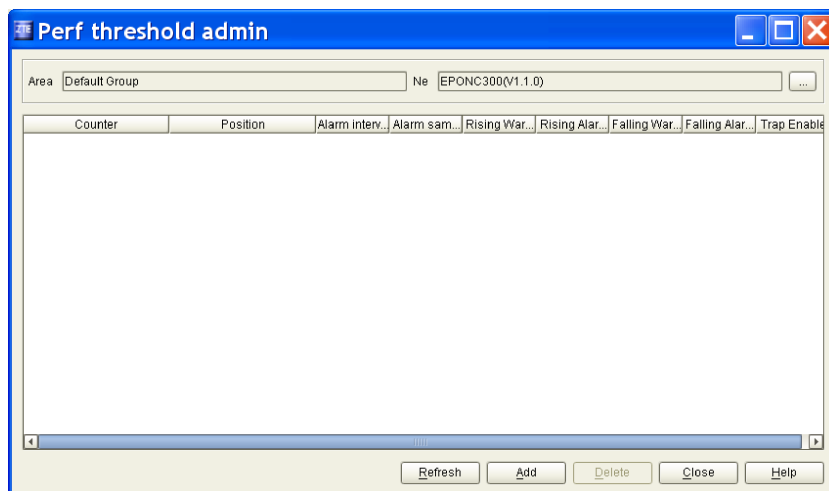
Short Description Perform this procedure to set the alarm threshold.

Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context To set the alarm threshold, perform the following steps:

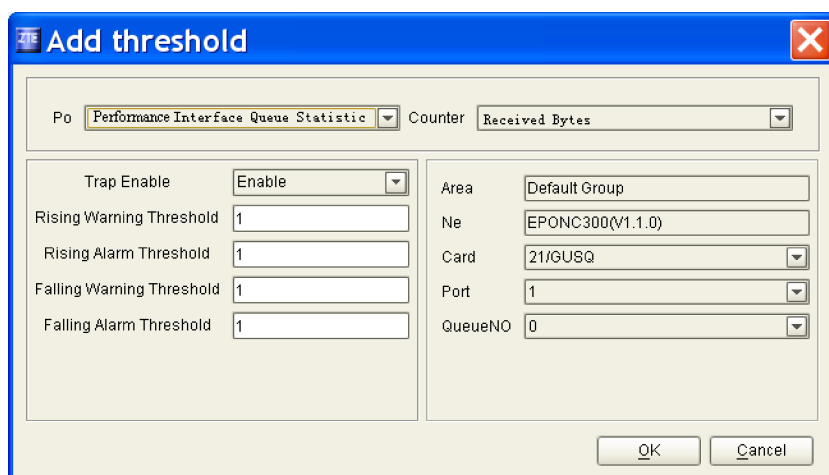
- Steps**
1. Right-click the ZXA10 C300 [NE](#) to choose **System Management > Perf Alarm Management**. The **Perf threshold admin** interface appears, as shown in [Figure 9](#).

FIGURE 9 PERF THRESHOLD ADMIN



2. Click **Add**. The **Add Threshold** interface appears, as shown in [Figure 10](#).

FIGURE 10 ADD PERFORMANCE THRESHOLD



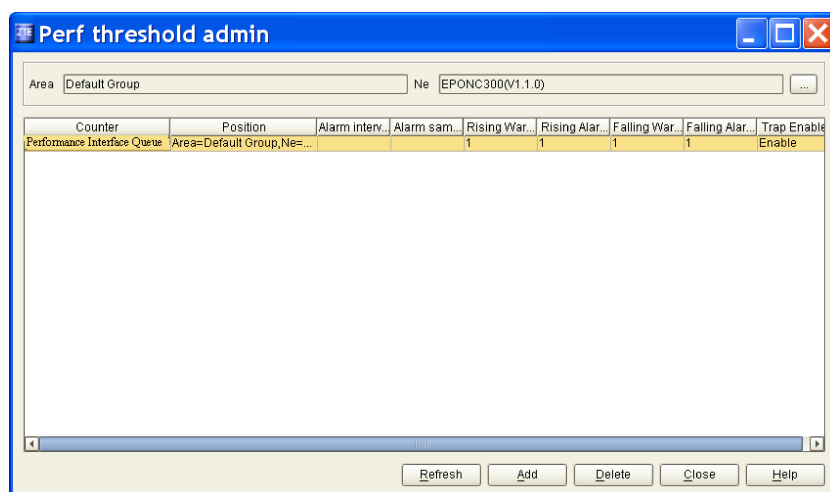
3. Set **Po** and **Counter**. The **Po** and **Counter** that can be configured are listed in [Table 5](#).

TABLE 5 PARAMETER DESCRIPTION FOR ADDED PERFORMANCE THRESHOLD

Performance Object	Counter
Ethernet Interface Queue Statistic	▶ Number of received bytes ▶ Number of received packets ▶ Number of sent bytes ▶ Number of sent packets ▶ Number of discarded bytes ▶ Number of discarded packets
MAC Address Table Usage Rate Statistic	▶ Maximum usage rate of MAC address table ▶ Minimum usage rate of MAC address table ▶ Average usage rate of MAC address table

4. Enter the ascending and descending warning threshold and alarm threshold. Click **OK**. The added performance threshold is displayed in the **Perf threshold admin** interface, as shown in [Figure 11](#).

FIGURE 11 PERF THRESHOLD ADMIN



END OF STEPS

Result The alarm threshold is set successfully.

Real-Time Monitoring

The NMS can monitor the alarms, events, and notifications in real time so that the users can know the changes and the faults occurring to the system in time. The NMS provides the following monitoring functions:

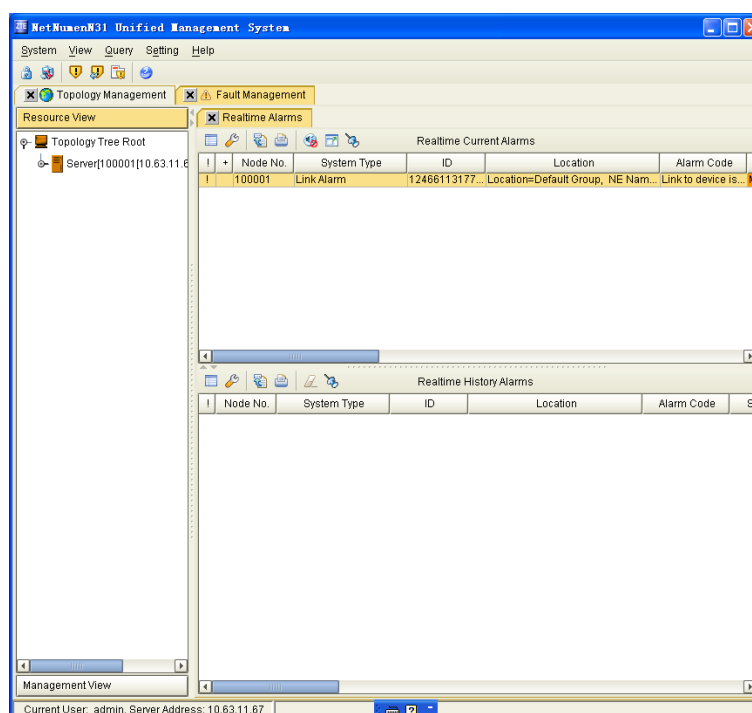
- Real-Time alarm monitoring
- Real-Time event monitoring
- Real-Time notification monitoring

After the NMS is started, the real-time monitoring function is enabled by default.

Real-Time Alarm Monitoring

Short Description	Perform this procedure to implement real-time alarms monitoring.
Prerequisites	The user has logged in to the NMS and the NMS manages ZXA10 C300 properly.
Context	The NMS supports real-time alarms monitoring so that the user can know the network running status in real time. When the system is faulty, the problem can be handled in time. Real-time alarm monitoring shows the real-time alarms that are not handled in the system. To implement the real-time alarm monitoring, perform the following steps:
Steps	1. On the NMS Client, choose View > Fault Management to open the Fault Management tab and the Realtime Alarms tab, as shown in Figure 12.

FIGURE 12 REAL-TIME ALARMS



2. The **Realtime Alarms** tab consists of two panes. The upper pane displays the current alarm monitoring, and the lower pane displays the history alarm monitoring.

END OF STEPS

Result The real-time alarm monitoring is complete.

Real-Time Event Monitoring

Short Description Perform this procedure to implement events real-time monitoring.

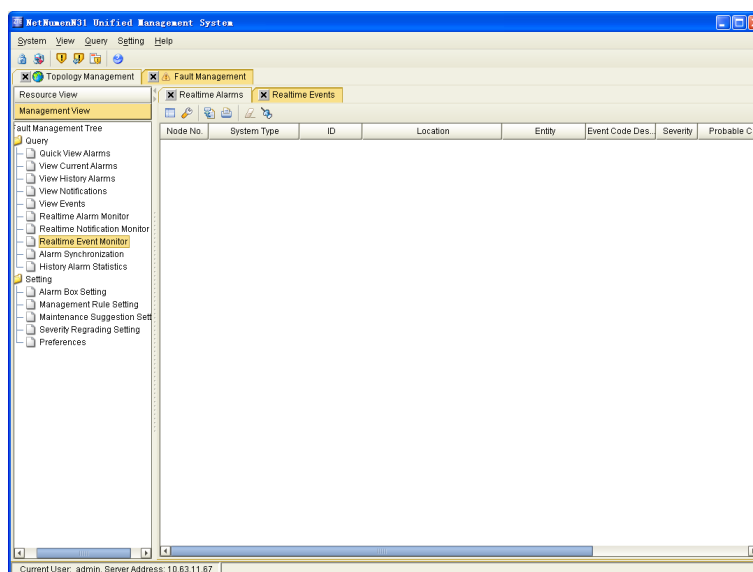
Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZX A10 C300 properly.

Context The NMS supports event real-time monitoring so that the user can know the network running status in real time.

To monitor the real-time events, perform the following steps:

- Steps**
1. On the NMS Client, choose **View > Fault Management** to open the **Fault Management** tab.
 2. Choose **Management View > Realtime Event Monitor** to open the **Realtime Events** tab, as shown in [Figure 13](#). The **Realtime Events** tab displays only the events that occur after the tab is opened.

FIGURE 13 REAL-TIME EVENTS

**END OF STEPS**

Result The real-time events monitoring is complete.

Real-Time Notification Monitoring

Short Description Perform this procedure to implement real-time notifications monitoring.

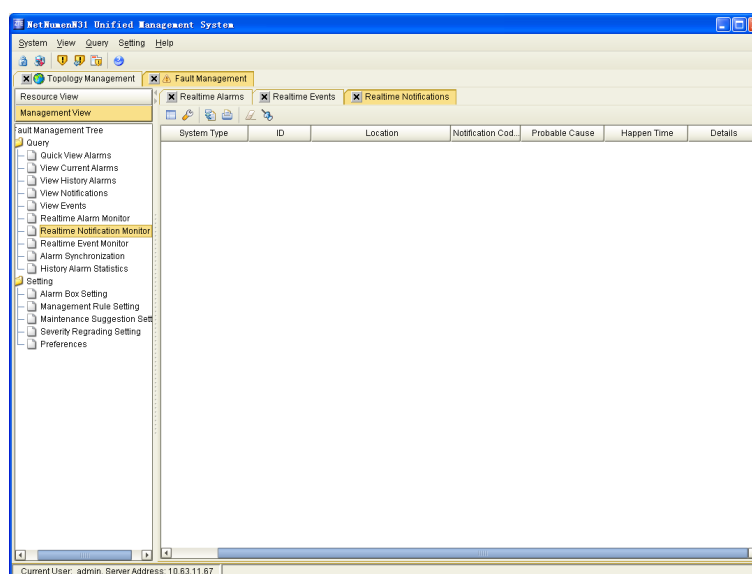
Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context The NMS supports real-time notification monitoring so that the user can know the network running status in real time.

To implement the real-time notification monitoring, perform the following steps:

- Steps**
1. On the NMS Client, choose **View > Fault Management** to open the **Fault Management** tab.
 2. Choose **Management View > Realtime Notification Monitor** to open the **Realtime Notifications** tab, as shown in [Figure 14](#). The **Realtime Notifications** tab displays only the notifications that occur after the tab is opened.

FIGURE 14 REAL-TIME NOTIFICATIONS

**END OF STEPS**

Result The real-time notification monitoring is complete.

Alarm Information Query

In the alarm management window, the user can query all the alarm data, such as the current alarms, history alarms, events, and notifications. When querying alarms, the user can set query conditions, such as the alarm occurrence location, alarm severity, possible cause, alarm system type, alarm description, occurrence time, alarm type, and alarm status. The query result is displayed in a list in the alarm window. The user can print or export the result to the local computer.

This topic includes the following:

- Querying Current Alarms
- Querying History Alarms
- Collecting History Alarm Statistics
- Querying Events
- Querying Notifications

Querying Current Alarms

Short Description Perform this procedure to query the current alarms.

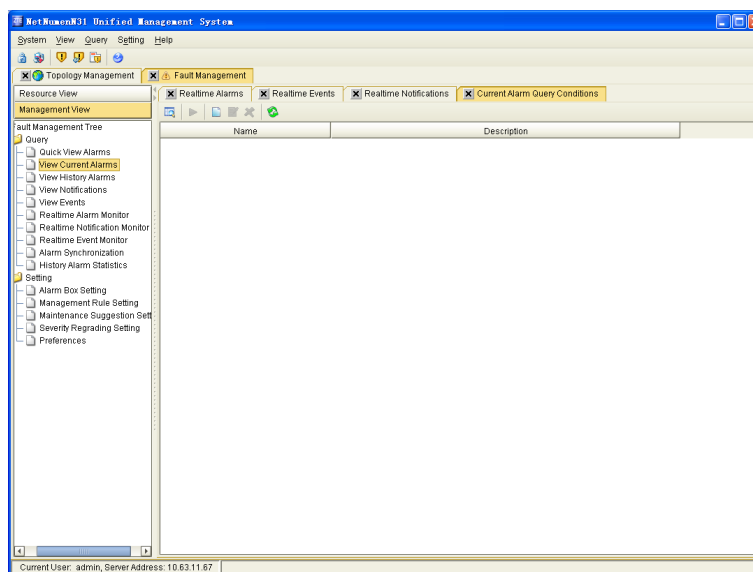
Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context The user can query the current alarms based on certain conditions on the NMS. The user can set the query conditions, such as the specific location, time, alarm type, alarm system type, and alarm cause.

To query the current alarms, perform the following steps:

- Steps**
1. On the NMS Client, choose **View > Fault Management** to open the **Fault Management** tab.
 2. Choose **Management View > View Current Alarms** to open the **Current Alarm Query Conditions** tab, as shown in [Figure 15](#).

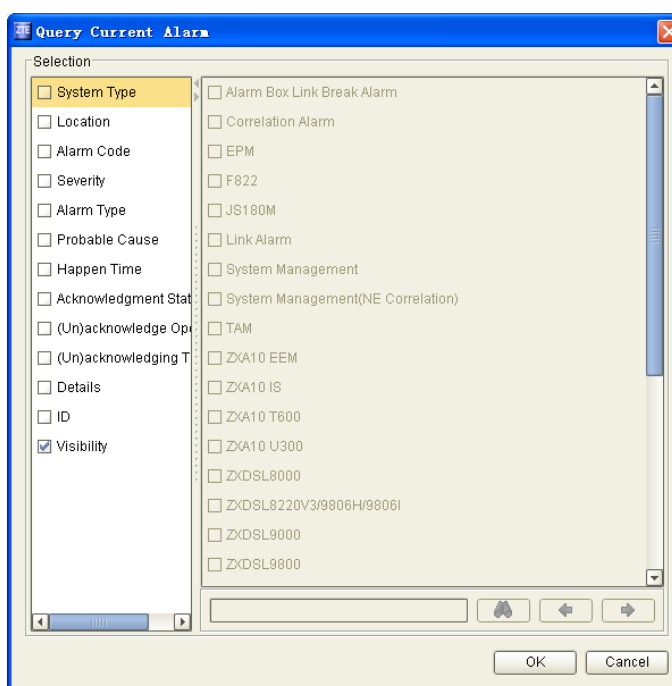
FIGURE 15 CURRENT ALARM QUERY CONDITIONS



3. On the **Current Alarm Query Conditions** tab, click the **Quick**

Search button  on the tool bar to display the **Quick Current Alarm** dialog box, as shown in [Figure 16](#).

FIGURE 16 QUERY CURRENT ALARM



4. In the **Query Current Alarm** dialog box, select the query condition. For example, select ZXA10 C300 (v1.0) for **System Type**. Then, click **OK** to view the query result.

END OF STEPS

Result Current alarm query is complete.

Querying History Alarms

Short Description Perform this procedure to query the history alarms.

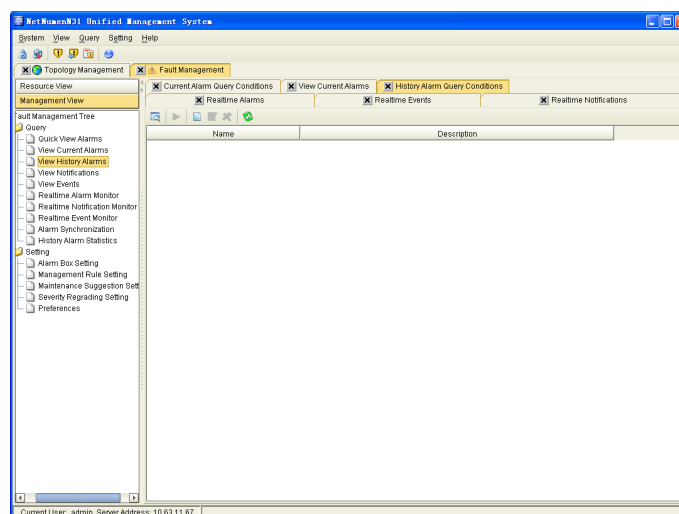
Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context The user can query the history alarms based on certain conditions on the NMS. The user can set the query conditions, such as the specific location, time, alarm type, alarm system type, and alarm cause.

To query the history alarms, perform the following steps:

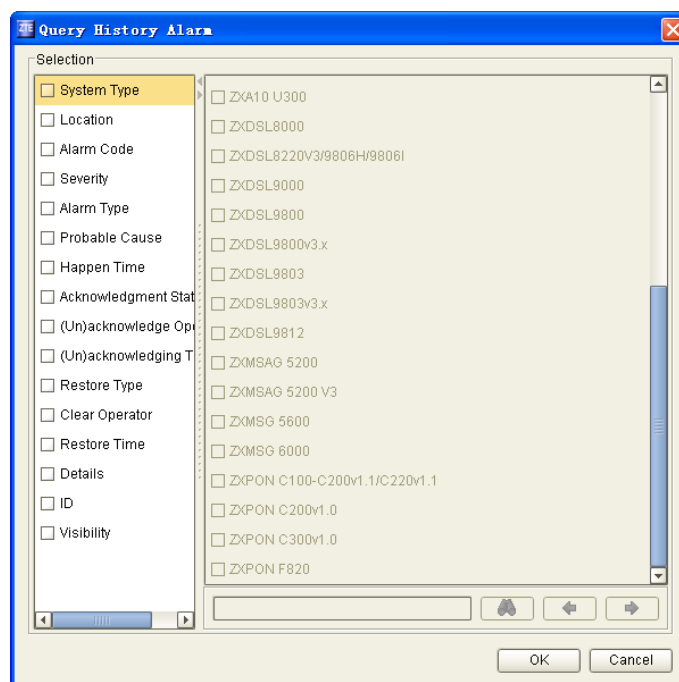
- Steps**
1. On the NMS Client, choose **View > Fault Management** to open the **Fault Management** tab.
 2. On the **Fault Management** tab, choose **Management View > View History Alarms** to open the **History Alarm Query Conditions** tab, as shown in [Figure 17](#).

FIGURE 17 HISTORY ALARM QUERY CONDITIONS



3. On the **History Alarm Query History Conditions** tab, click the **Quick Search** button  on the tool bar to open the **Query History Alarm** dialog box, as shown in [Figure 18](#).

FIGURE 18 QUERY HISTORY ALARM



4. In the **Query History Alarm** dialog box, select the querying condition. For example, select ZXA10 C300 (v1.0) for **System Type**. Then, click **OK** to view the query result.

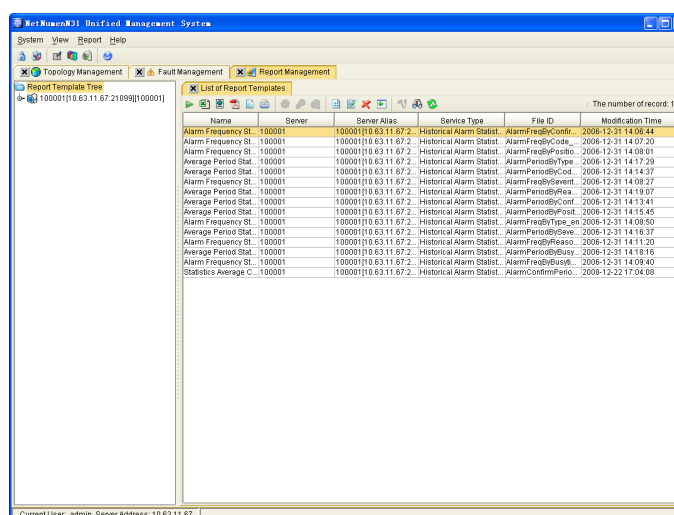
END OF STEPS

Result History alarm query is complete.

Collecting History Alarm Statistics

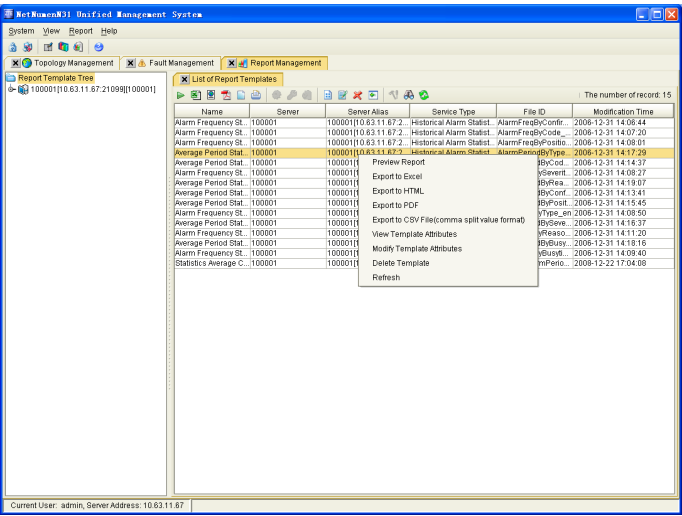
- Short Description** Perform this procedure to collect history alarm statistics.
- Prerequisites** The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.
- Context** The user can use the alarm statistics template of the NMS to collect statistics on the alarm data and generate reports. The user can export these reports in different formats or print them.
- To collect history alarm statistics, perform the following steps:
- Steps**
1. On the NMS Client, choose **View > Fault Management** to open the **Fault Management** tab.
 2. On the **Fault Management** tab, choose **Management View > History Alarm Statistics** to open the **Report Management** tab and the **List of Report Templates** tab, as shown in [Figure 19](#).

FIGURE 19 HISTORY ALARM STATISTICS — REPORT MANAGEMENT



3. Right-click a line and choose **Preview Report** or **Export to Excel**, as shown in [Figure 20](#).

FIGURE 20 HISTORY ALARM STATISTICS — LIST OF REPORT TEMPLATES



END OF STEPS

Result History alarm statistics is complete.

Querying Events

Short Description Perform this procedure to query events.

Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context The user can query events based on certain conditions on the NMS. The user can set the query conditions, such as the specific location, time, level, fault object, and event name.

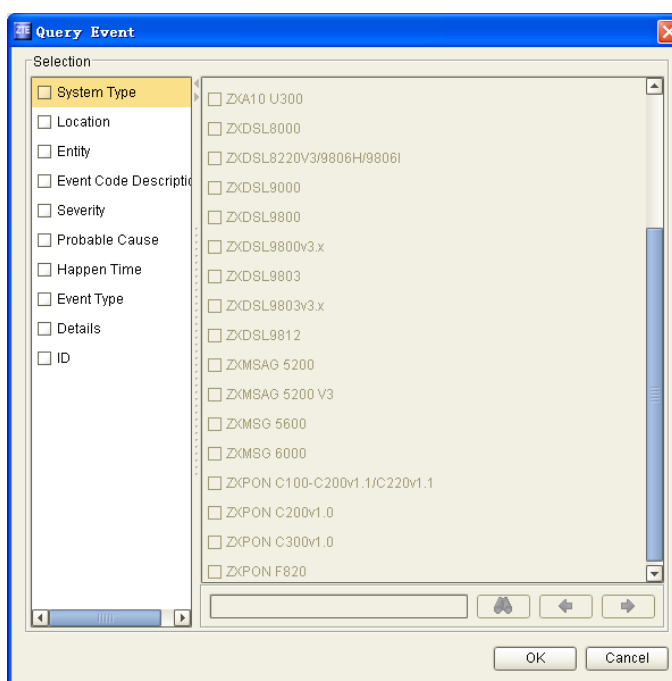
To query events, perform the following steps:

- Steps**
1. On the NMS Client, click **View > Fault Management** to open the **Fault Management** tab.
 2. On the **Fault Management** tab, choose **Management View > View Events** to open the **Event Query Conditions** tab.
 3. On the **Event Query Conditions** tab, click the **Quick Search**



button on the tool bar to display the **Query Event** dialog box, as shown in [Figure 21](#).

FIGURE 21 QUERY EVENT



4. In the **Query Event** dialog box, select the querying conditions. For example, select ZXA10 C300 (v1.0) for **System Type**. Then, click **OK** to view the querying result.

END OF STEPS

Result Event query is complete.

Querying Notifications

Short Description Perform this procedure to query notifications.

Prerequisites The user has logged in to the [NMS](#) and the NMS manages ZXA10 C300 properly.

Context The user can query notifications based on certain conditions on the NMS. The user can set the query conditions, such as the specific location, time, cause, description, and system type.

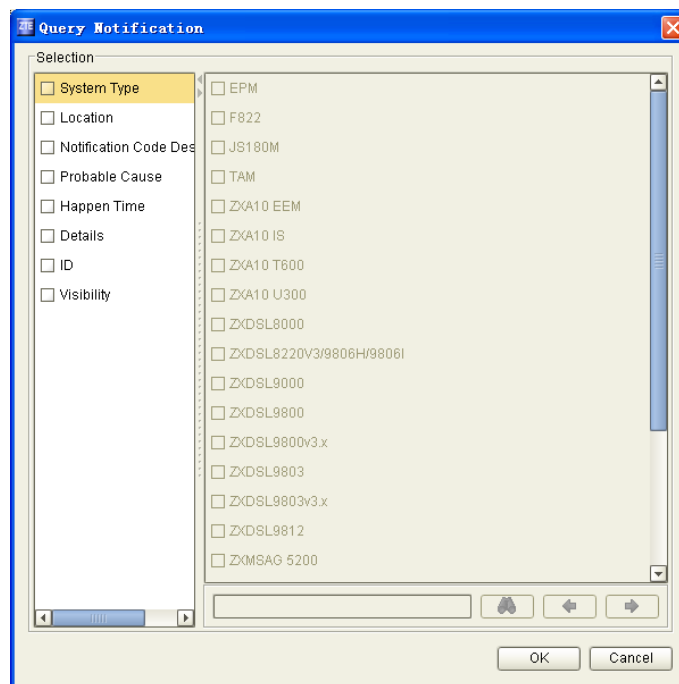
To query notifications, perform the following steps:

- Steps**
1. On the NMS Client, choose **View > Fault Management** to open the **Fault Management** tab.
 2. On the **Fault Management** tab, choose **Management View > View Notification** to open the **Notification Query Conditions** tab.
 3. On the **Notification Query Conditions** tab, click the **Quick**



Search button on the tool bar to display the **Query Notification** dialog box, as shown in [Figure 22](#).

FIGURE 22 QUERY NOTIFICATION



4. In the **Query Notification** dialog box, select the querying conditions. For example, choose the ZXA10 C300 (v1.0) for **System Type**. Then click **OK** to view the query result.

END OF STEPS

Result Notification query is complete.

Chapter 2

EPON Service Alarms

Table of Contents

Device OLT PortBer State.....	25
PON LOS	26
ONU Errored Symbol Period Event.....	26
ONU Errored Frame Period Event	27
ONU Errored Frame Period Event	28
ONU Errored Frame Seconds Summary Event.....	28
ONU Uplink Bit Error	29
ONU Uplink Frame Error	29
ONU Optical Power Threshold Crossing	30
ONU Offline	30
ONU Dying Gasp	31
OLT Reset.....	31
Local ONU Configure Failed (OLT)	32
Extended OAM	32
Extended OAM Discovery Failed (ONU)	35
Unknown ONU Online.....	36
Unknown ONU Offline (ONU).....	36
Deny Unknown ONU Register (ONU).....	36
ONU Request Auto Config	37

Device OLT PortBer State

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43014	Critical	Device alarm

Impact on System

Error bits exist on the physical link between the **OLT** and **ONU**. Packet loss may occur.

Possible Causes

The optical power received at the OLT side is too high or too low.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the optical fiber connection between the ZXA10 C300 device and the ONU is correct. If not, connect the optical fiber well.
2. Check whether the optical output power of the optical module on the **PON** card is normal. If not, replace optical module or the PON card.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

PON LOS

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43015	Critical	Device alarm

Impact on System All the **ONUs** under this **OLT** cannot work properly.

Possible Causes The fault may be caused by the following:

- The primary optical fiber is not well inserted.
- The primary optical fiber is broken.
- The OLT optical module is faulty.
- All the ONUs are powered OFF.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the primary optical fiber between the ZXA10 C300 device and ONU is loose or has poor contact. If yes, insert the optical fiber well.
2. Check whether the power is cut off in the area where ONU resides. Restore the mains supply.
3. Check whether the optical output power of the optical module on the **PON** card is normal. If not, replace the optical module or PON card.
4. Check whether the primary optical fiber between the ZXA10 C300 device and ONU is broken. Make sure that the optical fiber is connected well.
5. If the alarm still exists, contact technical support personnel for assistance.

Result The alarm disappears.

ONU Errored Symbol Period Event

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43017	Critical	Device alarm

Impact on System It may lead to packet loss on link or cause broken link.

Possible Causes The OLT detects the number of the downstream error notations at the physical layer of the ONU in a period of time. When the number exceeds the threshold, the system generates the alarm.

- Troubleshooting** To handle the fault, perform the following steps:
1. Check whether the optical fiber between the device and ONU bends too much, or whether the optical fiber is well inserted. If yes, rearrange and insert the optical fiber well.
 2. Check whether the attenuator is suitable. If not, replace the attenuator.
 3. Check whether the optical fiber and optical module are aging. If yes, replace the optical fiber and optical module.
 4. If the alarm still exists, contact the technical support personnel for assistance.
- Result**
The alarm disappears.

ONU Errored Frame Period Event

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43019	Critical	Device alarm

Impact on System It may lead to packet loss on link or cause broken link.

Possible Causes Check the number of the downstream error frames received at the **ONU MAC** sub-layer in a period of time. When the number exceeds the threshold, the system generates the alarm.

- Troubleshooting** To handle the fault, perform the following steps:
1. Check whether the optical fiber between the device and ONU bends too much, or whether the optical fiber is well inserted. If yes, rearrange and insert the optical fiber well.
 2. Check whether the attenuator is suitable. If not, replace the attenuator.
 3. Check whether the optical fiber and optical module are aging. If yes, replace the optical fiber and optical module.
 4. If the alarm still exists, contact the technical support personnel for assistance.
- Result** The alarm disappears.

ONU Errored Frame Period Event

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43018	Critical	Device alarm

Impact on System

It may lead to packet loss on link or cause broken link.

Possible Causes

Check the number of the downstream error frames at the sub-layer of **ONU MAC** in a period of time. When the number exceeds the threshold, the system generates the alarm.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the optical fiber between the device and ONU bends too much, or whether the optical fiber is well inserted. If yes, rearrange and insert the optical fiber well.
2. Check whether the attenuator is suitable. If not, replace the attenuator.
3. Check whether the optical fiber and optical module are aging. If yes, replace the optical fiber and optical module.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result

The alarm disappears.

ONU Errored Frame Seconds Summary Event

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43020	Critical	Device alarm

Impact on System

It may lead to packet loss on link or cause broken link.

Possible Causes

Check the number of the downstream error frame seconds at the sub-layer of **ONU MAC** in a period of time. When the number exceeds the threshold, the system generates the alarm.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the optical fiber between the device and ONU bends too much, or whether the optical fiber is well inserted. If yes, rearrange and insert the optical fiber well.
2. Check whether the attenuator is suitable. If not, replace the attenuator.
3. Check whether the optical fiber and optical module are aging. If yes, replace the optical fiber and optical module.

4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

ONU Uplink Bit Error

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43021	Critical	Device alarm

Impact on System It may lead to packet loss on link or cause broken link.

Possible Causes The ONU upstream error bit rate exceeds the threshold.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the optical fiber between the ONU uplink and PON port is well inserted or connected improperly. If not, reinsert the optical fiber.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

ONU Uplink Frame Error

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43022	Critical	Device alarm

Impact on System It may lead to packet loss on link or cause broken link.

Possible Causes The ONU upstream error bit rate or error frame number exceeds the threshold. By default, the threshold of the error bit rate is 1×10^{-10} , and the minimum threshold of the error frame number is 1000.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the optical fiber between the ONU uplink and PON port is well inserted. If not, insert it well.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

ONU Optical Power Threshold Crossing

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43009	Critical	Device alarm

Impact on System This alarm may lead to error bit or packet loss. In serious cases, it may lead to **ONU** disconnection.

Possible Causes When the ONU optical power received on the **OLT** is too high or too low, the system generates the alarm.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the optical fiber distance meets the standard requirements. If not, replace the attenuator or adjust the optical fiber distance to meet the standard requirements.
2. Check whether the optical fiber bends too much. If yes, rearrange or replace the optical fiber.
3. Check whether the ONU optical module are aging. If yes, replace the optical module.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

ONU Offline

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43016	Critical	Device alarm

Impact on System The **OLT** may fail to manage **ONU**. The service is broken.

Possible Causes The fault may be caused by the following:

- The ONU link is down.
- A branch optical fiber is broken.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the fiber connecting the ONU is pulled out or improperly inserted. If yes, reinsert the fiber.
2. Check whether the optical fiber between the ONU uplink and **PON** port is faulty. If yes, rearrange the fiber or replace the fiber.
3. Check whether the ONU power module is damaged. If yes, replace the power module or the whole ONU.

4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

ONU Dying Gasp

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43061	Warning	Communication alarm

Impact on System The OLT may fail to manage ONU. The service is broken.

Possible Causes The fault may be caused by the following:

- The power supply is cut off in the area where the ONU resides.
- The ONU voltage is not stable or the power module is damaged.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the fiber connecting the ONU is pulled out or improperly inserted. If yes, reinsert the fiber.
2. Check whether the optical fiber between the ONU uplink and PON port is faulty. If yes, rearrange the fiber or replace the fiber.
3. Check whether the ONU power is OFF. If the alarm is caused by power failure, restore the power supply.
4. Check whether the ONU power module is damaged. If yes, replace the power module or the whole ONU.
5. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

OLT Reset

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43049	Notification	Communication alarm

Impact on System The ONU is disconnected during initiation and the service is not available.

Possible Causes The OLT is restarted.

Troubleshooting No need to handle it. The service recovers after the ONU is restarted.

Result The alarm disappears.

Local ONU Configure Failed (OLT)

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43053	Notification	Communication alarm

Impact on System Some services are unavailable.

Possible Causes When some properties, such as the [ONU](#) bandwidth and encryption, fail to be configured, the system generates the alarm.

Troubleshooting To handle the fault, perform the following steps:

1. Re-configure the ONU.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Extended OAM

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43060	Notification	Communication alarm

Impact on System The [ONU](#) fails to work properly or some services are unavailable.

Possible Causes The important parameters of ONU optical transceiver exceed the alarm threshold and the ONU reports to [OLT](#) through the extended [OAM](#) channel in transparent mode. [Table 6](#) provides the details.

TABLE 6 FAULT ANALYSIS FOR ONU OPTICAL MODULE DIAGNOSTIC PARAMETER ALARM

Possible Causes	Fault Analysis
Port (Up)	The ONU user port is up.
Port (Down)	The ONU user port is down.
Auto-negotiation failure	Auto-negotiation fails on the user port.
Insert user line	Connect the VoIP user phone set.
Pull out user line	Remove the VoIP user phone set.
User hook-off	The VoIP user phone set is hooked off.
User hook-on	The VoIP user phone set is hooked on.
User line is faulty	The VoIP user line is faulty.

Possible Causes	Fault Analysis
E1 LOS	PDH E1 signal loss occurs.
E1 LOF	PDH E1 frame loss occurs.
E1 AIS	PDH E1 alarm indicating signal is generated.
Clock is not locked.	The PDH clock is not locked.
Clock is not locked.	The PDH clock is not locked.
The working temperature of ONU optical module is normal.	The ONU temperature is adjusted appropriately.
The working temperature of ONU optical module is higher than the high alarm threshold.	The ONU detects that the working temperature of the optical module is higher than the configured alarm threshold, which leads to ONU disconnection after some time.
The working temperature of ONU optical module is lower than the low alarm threshold.	The ONU detects that the working temperature of the optical module is lower than the configured alarm threshold, which leads to ONU disconnection after some time.
The working temperature of ONU optical module is higher than the high alarm threshold.	The ONU detects that the working temperature of the optical module is higher than the alarm threshold. It is a reminder, implying that the ONU temperature is a little higher.
The working temperature of ONU optical module is higher than the high alarm threshold.	The ONU detects that the working temperature of the optical module is higher than the alarm threshold. It is a reminder, implying that the ONU temperature is a little higher.
The voltage of ONU optical module is normal.	A normal notification is raised.
The working temperature of ONU optical module is higher than the high alarm threshold.	The ONU detects that the voltage of the ONU optical module is higher than the high alarm threshold.
The voltage of ONU optical module is lower than the low alarm threshold.	The ONU detects that the voltage of the ONU optical module is lower than the low alarm threshold.
The voltage of ONU optical module is higher than the high alarm threshold.	The ONU detects that the voltage of the ONU optical module is higher than the high alarm threshold.

Possible Causes	Fault Analysis
The voltage of ONU optical module is lower than the low alarm threshold.	The ONU detects that the voltage of the ONU optical module is higher than the high alarm threshold.
The bias current of ONU optical module is normal.	The ONU detects that the bias current of the ONU optical module is normal.
The bias current of ONU optical module is higher than the high alarm threshold.	The ONU detects that the bias current of the ONU optical module is higher than the high alarm threshold.
The bias current of ONU optical module is lower than the low alarm threshold.	The ONU detects that the bias current of the ONU optical module is lower than the low alarm threshold.
The bias current of ONU optical module is lower than the low alarm threshold.	The ONU detects that the bias current of the ONU optical module is lower than the low alarm threshold.
The bias current of ONU optical module is lower than the low alarm threshold.	The ONU detects that the bias current of the ONU optical module is lower than the low alarm threshold.
ONU optical module sends optical power normally.	The ONU detects that ONU optical module sends optical power normally.
The optical power that ONU optical module sends is higher than the high alarm threshold.	The ONU detects that the optical output power of the ONU optical module is higher than the high alarm threshold.
The optical power that ONU optical module sends is lower than the low alarm threshold.	The ONU detects that the optical output power of the ONU optical module is lower than the low alarm threshold. It may cause the ONU registration failure. If the current, voltage and temperature are normal, the optical module is aging.
The optical power that ONU optical module sends is higher than the high alarm threshold.	The ONU detects that the optical output power of the ONU optical module is higher than the high alarm threshold.
The optical power that ONU optical module sends is higher than the low alarm threshold.	The ONU detects that the optical output power of the ONU optical module is higher than the low alarm threshold.
ONU optical module sends optical power normally.	A normal notification is raised.

Possible Causes	Fault Analysis
The optical power that ONU optical module sends is higher than the high alarm threshold.	When the ONU detects that the optical input power of the optical module is higher than the high alarm threshold, increase the optical distance or increase the attenuation.
The optical power that ONU optical module receives is lower than the low alarm threshold.	When ONU detects that the optical input power of the optical module is lower than the low alarm threshold, shorten the optical distance or decrease the attenuation.
The optical power that ONU optical module receives is higher than the high alarm threshold.	ONU detects that the optical input power of the optical module is higher than the high alarm threshold.
The optical power that ONU optical module receives is higher than the low alarm threshold.	ONU detects that the optical input power of the optical module is higher than the low alarm threshold.
Low battery voltage	The battery may have aged. Replace the battery.

Troubleshooting Refer to "Possible Causes".

Result The alarm disappears.

Extended OAM Discovery Failed (ONU)

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43052	Notification	Communication alarm

Impact on System The OLT may fail to manage the ONU. The service is broken.

Possible Causes The ONU is not supported or the ONU works improperly.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the connected ONU is supported.
 - ▶ If the ONU type is supported, the ONU may be abnormal. Repair the ONU or replace the ONU.
 - ▶ If the ONU type is not supported, replace the ONU type.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Unknown ONU Online

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43058	Notification	Communication alarm

Impact on System None**Possible Causes** An illegal ONU is connected.**Troubleshooting** Check whether the ONU is illegal. If yes, contact the technical support personnel for assistance.**Result** The alarm disappears.

Unknown ONU Offline (ONU)

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43054	Notification	Communication alarm

Impact on System None**Possible Causes** An illegal ONU is disconnected.**Troubleshooting** Check whether the ONU is illegal. If yes, contact the technical support personnel for assistance.**Result** The alarm disappears.

Deny Unknown ONU Register (ONU)

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43055	Notification	Communication alarm

Impact on System None**Possible Causes** Illegal ONU registration fails.**Troubleshooting** Check whether the ONU is illegal. If yes, contact the technical support personnel for assistance.**Result** The alarm disappears.

ONU Request Auto Config

Alarm Property

Alarm ID	Alarm Level	Alarm Type
43062	Notification	Communication alarm

Impact on System None

Possible Causes The auto-configuration request is enabled. When an unknown ONU is online, the NM configures the ONU with the default template.

Troubleshooting None

Result The alarm disappears.

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Chapter 3

CES Service Alarms

Table of Contents

PDH Interface LOS Alarm.....	39
PDH Interface AIS Alarm	40
PW Local Alarm	41
PW Remote Alarm.....	41
Optical Interface LOS Alarm	42
Loss of Frame Alarm	42
Out of Frame Alarm	43
MS-AIS	44
MS-RDI	44
RS SF	45
RS SD	45
AU-LOP	46
AU AIS	46
HP TIM	47
HP UNEQ	47
HPRDI	48
HP PLM	49
HP-AIS	49
HP TU LOM	50
TU-LOP	50
TU LOM	51
TU-AIS	51
LP TIM	52
LP UNEQ	52
LPRDI	53
LP PLM	54

PDH Interface LOS Alarm

Alarm Description When the signal disappears on the [PDH](#) interface of the circuit simulated card, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36865	Critical	Fault alarm

Impact on System It leads to CESoP service interruption.

Possible Causes

The fault may be caused by the following:

- The PDH cable is incorrectly connected.
- The opposite-end device does not send electrical signals.

- Troubleshooting** To handle the fault, perform the following steps:
1. Check whether ZXA10 C300 is well connected with the peripheral cables. If not, reconnect the cables.
 2. Check whether the hardware is faulty. If the circuit simulated card, such as CTBB or CTTB, is faulty, replace the card and check whether the system generates the **LOS** recovery alarm on the PDH interface.
 3. Check whether the opposite-end device sends signal normally. If not, handle the fault on the opposite-end device.
 4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

PDH Interface AIS Alarm

Alarm Description When the **PDH** interface on a circuit simulated card receives the **AIS** alarm signal, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36866	Critical	Fault alarm

Impact on System It leads to CESoP service interruption.

Possible Causes The fault may be caused by the following:

- The PDH cable is incorrectly connected.
- The opposite-end device is faulty.

- Troubleshooting** To handle the fault, perform the following steps:
- Carry out the local loopback command to perform near-end loopback or use cables to perform physical loopback. Check whether the alarm disappears to judge whether the card is faulty.
 - If the circuit simulated card, such as CTBB or CTTB, is faulty, replace the card. Check whether the system generates the AIS recovery alarm on the PDH interface.
 - Check whether ZXA10 C300 is well connected with peripheral cables. If not, reconnect the cables.
 - Check whether the opposite-end device communicates normally.
 - If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

PW Local Alarm

Alarm Description It is a local alarm. When the system detects that the local upstream **TDM** circuit is faulty, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36867	Major	Fault alarm

Impact on System It may lead to CESoP services interruption.

Possible Causes The **PDH** signal has the **LOS** alarm or **AIS** alarm.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether ZXA10 C300 is well connected with the opposite-end device. If not, reconnect the cables and check whether the system generates the **PW** local recovery alarm.
2. Check whether the configuration on ZXA10 C300 and CESoP services is correct. If not, re-configure the related data.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

PW Remote Alarm

Alarm Description It is a remote alarm. When the lost packets in the receiving direction at the remote end exceed the specified percentage in a certain period, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36868	Critical	Fault alarm

Impact on System It may lead to CESoP service interruption.

Possible Causes The receiving port at the opposite network side has packet loss status.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether ZXA10 C300 is well connected with the opposite-end device. If not, reconnect the cable and check whether the system generates the **PW** recovery alarm.
2. Check whether the configuration on ZXA10 C300 and CESoP services is correct. If not, re-configure the related data.
3. Check whether the opposite-end device works properly. If not, handle the device fault.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Optical Interface LOS Alarm

Alarm Description When the system detects that SDH-layer LOS occurs on the STM-1 or STM-4 optical interface of the CTLA card, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36869	Critical	Fault alarm

Impact on System If protection is configured, service protection is enabled and the services are transferred to the protected optical interface. Otherwise, service interruption is caused.

Possible Causes The fault may be caused by the following:

- The optical line is faulty.
- The optical module is faulty.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the optical fiber is well inserted.
2. Check whether the optical fiber is bent too much, or the joint is loose or the joint ends attenuate too much. If the optical fiber is faulty, replace it.
3. Check whether the system optical output and input power is proper.
4. Check whether the hardware is faulty. If the card is faulty, replace the card. If the receiving port is faulty, replace its optical card. If the transmission port is faulty, replace its optical card. After replacement, check whether the system generates the LOS recovery alarm.
5. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Loss of Frame Alarm

Alarm Description When the system detects that SDH-layer LOF occurs on the STM-1 or STM-4 port of the CTLA card, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36870	Critical	Fault alarm

Impact on System If protection is configured, service protection is enabled and the services are transferred to the protected optical interface. Otherwise, service interruption is caused.

Possible Causes The fault may be caused by the following:

- The optical line card is faulty.

- The error bits on the optical line are severe.
- The optical port rates on the transmitting port and receiving port do not match.

Troubleshooting To handle the fault, perform the following steps:

1. Reboot the main control card or pull out and insert the main control card manually. Check whether the LOF alarm is reported.
2. Check whether the rates on the transmitting port and receiving port match. Set the rates to be consistent.
3. Carry out the local loopback command to perform loopback. If the conditions allow, use optical fiber to perform physical loopback. Check whether the [NE](#) at the receiving side or the transmission side is faulty. Replace the optical line card or the main control card where the NE is faulty.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Out of Frame Alarm

Alarm Description When the system detects that [SDH](#)-layer Out-of-frame (OoF) error occurs on the [STM](#)-1 or STM-4 port of the CTLA card, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36871	Critical	Fault alarm

Impact on System It may lead to service bit error or service interruption.

Possible Causes The fault may be caused by the following:

- The optical line card is faulty.
- The error bits on the optical line are severe.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the optical line is normal. Fix the optical line and ensure normal communication. Check whether the OoF recovery alarm is reported.
2. Carry out the local loopback command to perform local loopback. If conditions allow, use the optical fiber to perform physical loopback. Check whether the [NE](#) is faulty at the receiving side or the transmission side. Replace the optical line or the main control card where the faulty NE exists.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

MS-AIS

Alarm Description When the system receives [MS-AIS](#), it generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36872	Critical	Fault alarm

Impact on System It may lead to service interruption or enabling service protection.

Possible Causes MS-AIS bits are all 1.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether MS-AIS is reported to the uplink device or the rear three bits of K2 are set to be 1. If yes, handle the uplink fault.
2. Check whether the signal is transmitted properly from the uplink. Make sure that the upstream device works properly.
3. If some other alarms (either remote [LOS/LOF/AIS](#)/LoP or local [RDI](#)) occur simultaneously, check whether the corresponding maintenance is implemented at the remote or local end.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

MS-RDI

Alarm Description When the system receives [MS-RDI](#), the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36873	Critical	Fault alarm

Impact on System There is no effect on the services of the receiving party.

Possible Causes The opposite-end [NE](#) sends back the alarm after detecting the re-generation section alarm or [MS-AIS](#) alarm.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the local device reports the [LOS](#) or MS-AIS alarm to the uplink device or whether the rear three bits of K2 are set to be 1. If yes, handle the uplink fault. Check whether the MS-RDI alarm is raised.
2. Check whether the local device sends signals normally. Make sure that the local device works properly.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

RS SF

Alarm Description When the multiplex section generates larger bit errors ($BER > 10e^{-3}$), the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36874	Minor	Fault Alarm

Impact on System It may lead to service bit error or service interruption.

Possible Causes The fault may be caused by the following:

- The optical line is faulty.
- The optical line is faulty.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the optical fiber is bent too much. Clean the fiber joints. Replace the optical fiber if necessary to make sure that the optical line attenuates normally. Check whether the RS SF recovery alarm is reported.
2. Check whether the optical line card on the receiving and the transmitting ports and the main control card work properly. If the cards are faulty, replace the cards.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

RS SD

Alarm Description When fewer bit errors are generated in the multiplex section, $BER > 10e^{-6}$, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36875	Minor	Fault alarm

Impact on System It may lead to service bit error or service interruption.

Possible Causes The fault may be caused by the following:

- The optical line is faulty.
- The optical line card is faulty.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the optical fiber is bent too much. Clean the joint ends. Replace the optical fiber if necessary. Ensure that the optical line attenuation is normal.
2. Check whether the optical line card and the main control card on the receiving and the transmitting ports are normal. Replace the cards if they are faulty.

3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

AU-LOP

Alarm Description When the system detects that SDH-layer AU-LOP occurs on the STM-1 or STM-4 port, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36876	Critical	Fault alarm

Impact on System It may lead to service interruption.

Possible Causes

The fault may be caused by the following:

- The AU signal source is faulty.
- The NE main control card in the upstream AU transmission direction is faulty.
- The local main control card is faulty.

Troubleshooting

To handle the fault, perform the following steps:

1. Ensure that the AU signal source is normal. Check whether the AU-LOP alarm is cleared.
2. Check whether the NE main control card in the upstream AU transmission direction runs normally. If not, replace the faulty card.
3. Check whether the local main control card runs properly. Replace the faulty card if necessary.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

AU AIS

Alarm Description When the system receives AU AIS, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36877	Critical	Fault alarm

Impact on System It may lead to service interruption.

Possible Causes

The fault may be caused by the following:

- The high-level time slot cross on the uplink node fails.

- Troubleshooting**
- The AU signal source is faulty.
- To handle the fault, perform the following steps:
1. Ensure that the AU signal source is normal. Check whether the AU AIS alarm is cleared.
 2. Check whether the high-level time slot cross on the uplink node is successful (by viewing the protection configuration library and time slot configuration library). Make sure that the upstream device works properly.
 3. If the alarm still exists, contact the technical support personnel for assistance.
- Result** The alarm disappears.

HP TIM

Alarm Description When the system detects that HP TIM occurs, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36878	Minor	Fault alarm

Impact on System None

Possible Causes The fault may be caused by the following:

- The mode configuration on the local device is inconsistent with that on the opposite device.
- The expected value of the channel ID (J1) set on the local device is inconsistent with the transmission value set on the opposite device.

- Troubleshooting**
- To handle the fault, perform the following steps:
1. Check whether the mode configuration on the local device is consistent with that on the opposite device. Make sure that they are consistent. Check whether the HP TIM alarm is cleared.
 2. Check whether the expected value of the channel ID (J1) set on the local device is consistent with the transmission value set on the opposite device. Make sure that they are consistent.
 3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

HP UNEQ

Alarm Description When the system detects HP UNEQ, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36879	Critical	Fault alarm

Impact on System The service is not available on the channel.

Possible Causes

The fault may be caused by the following:

- The corresponding upstream channel is not properly configured.
- The high-level time slot cross on the uplink node fails.
- The configuration-type database is not downloaded properly.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the corresponding upstream channel is properly configured. Make sure that the upstream device works properly. Check whether the HP UNEQ alarm is cleared.
2. Check whether the high-level time slot cross on the uplink node is successful. Make sure that the upstream device works properly.
3. Check whether the configuration-type database is properly downloaded. If not, download the configuration-type database.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result

The alarm disappears.

HPRDI

Alarm Description

When the STM-1 or the STM-4 port receives HPRDI, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36880	Critical	Fault alarm

Impact on System None

Possible Causes

The fault may be caused by the following:

- When the uplink node receives an alarm, such as AU-LOP, AU-AIS, HP-UNEQ, HP-PLM, or HP-TIM, it sends the alarm back to the local device.
- The uplink node detects the AU-level critical alarm caused by the alarms on the multiplex section.
- The node optical signal or AU transmission is abnormal.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the uplink device receives the multiplex section or AU-level alarm. If yes, handle the fault.

2. Check whether the AU time slot cross on the local node is normal and is sent to the uplink node. Make sure that the local node works normally.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

HP PLM

Alarm Description When the system detects HP-PLM, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36881	Minor	Fault alarm

Impact on System It may lead to service interruption.

Possible Causes The expected value of the payload ID (C2) set on the local device is inconsistent with the transmission value set on the opposite device.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the expected value of the payload ID (C2) set on the local device is inconsistent with the transmission value on the opposite device. If not, keep them consistent. Check whether the HP PLM alarm is cleared.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

HP-AIS

Alarm Description When the system receives HP-AIS, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36882	Critical	Fault alarm

Impact on System It may lead to service interruption.

Possible Causes The fault may be caused by the following:

- The AU signal source is faulty.
- The high-level time slot cross on the uplink node fails.

Troubleshooting To handle the fault, perform the following steps:

1. Ensure that the AU signal source is normal. Check whether the HP-AIS alarm is cleared.
2. Check whether the high-level time slot cross on the uplink node is successful (by viewing the protection configuration library and time slot configuration library). Check whether the uplink device works normally.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

HP TU LOM

Alarm Description When the system detects HP LOM on the STM-1 or STM-4 port, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36883	Critical	Fault alarm

Impact on System The HP service is unavailable.

Possible Causes There may be consecutive bit errors.

Troubleshooting To handle the fault, perform the following steps:

1. Perform line loopback or the device loopback to check whether the local device or opposite device is faulty. Replace the faulty NE optical line card or the main control card.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

TU-LOP

Alarm Description When the system detects that TU-LOP occurs on the STM-1 or the STM-4 port, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36884	Critical	Fault alarm

Impact on System The service is not available on the channel.

Possible Causes The fault may be caused by the following:

- The main control card is improperly inserted.
- The main control card fails to work properly.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the main control card is properly inserted. If not, pull out the card and insert it again. Check whether the TU-LOP alarm is cleared.
2. Replace the main control card. Check whether the TU-LOP alarm is cleared.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

TU LOM

Alarm Description When the system detects that TU LOM (Loss of multiframe) occurs on the STM-1 or the STM-4 port, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36885	Critical	Fault alarm

Impact on System The service is unavailable on the channel.

Possible Causes The H4 pointer is lost.

Troubleshooting To handle the fault, perform the following steps:

1. Perform line loopback or device loopback to check whether the local device or opposite device is faulty. Replace the faulty NE optical line card or the main control card.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

TU-AIS

Alarm Description When the system receives TU-AIS, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36886	Critical	Fault alarm

Impact on System The service is unavailable on the channel.

Possible Causes The fault may be caused by the following:

- The TU signal source is faulty.
- The low-level time slot cross on the uplink node fails.

Troubleshooting To handle the fault, perform the following steps:

1. Make sure that the TU signal source is normal. Check whether the TU-AIS alarm is cleared.
2. Check whether the low-level time slot cross on the uplink node is successful (by viewing the protection configuration library and time slot configuration library). Make sure that the uplink device works normally.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

LP TIM

Alarm Description When the system detects that LP TIM occurs, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36887	Minor	Fault alarm

Impact on System None

Possible Causes

The fault may be caused by the following:

- The mode configuration on the local device is inconsistent with that on the opposite device.
- The expected value of the channel ID (J2) set on the local device is inconsistent with the transmission value set on the opposite device.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the mode configuration on the local device is consistent with that on the opposite device. Make sure that they are consistent. Check whether the LP TIM alarm is cleared.
2. Check whether the expected value of the channel ID (J2) set on the local device is consistent with the transmission value set on the opposite device. Make sure that they are consistent.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

LP UNEQ

Alarm Description When the system detects LP UNEQ, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36888	Critical	Fault alarm

Impact on System The service is unavailable on the channel.

Possible Causes The fault may be caused by the following:

- The corresponding upstream channel is not properly configured.
- The low-level time slot cross on the uplink node fails.
- The configuration-type database is not downloaded properly.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the corresponding upstream channel is properly configured. Make sure that the upstream device works properly. Check whether the LP UNEQ alarm is cleared.
2. Check whether the low-level time slot cross on the uplink node is successful. Make sure that the upstream device works properly.
3. Check whether the configuration-type database is properly downloaded. If not, download the configuration-type database.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

LPRDI

Alarm Description When the STM-1 or the STM-4 port receives LPRDI, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36889	Critical	Fault alarm

Impact on System None

Possible Causes The fault may be caused by the following:

- When the upstream node receives an alarm, such as TU-LOP, TU-AIS, LP-UNEQ, or LP-TIM, it sends the alarm back to the local device.
- The uplink node detects the TU-level critical alarm caused by the alarms on the high order path.
- The local node TU or AU transmission is abnormal.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the uplink device receives the HP or TU-level alarm. If yes, handle the fault.

2. Check whether the TU time slot cross on the local node is normal and is sent to the uplink node. Make sure that the local node works normally.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

LP PLM

Alarm Description When the system detects LP PLM, the system generates the alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
36890	Minor	Fault alarm

Impact on System It may lead to service interruption.

Possible Causes The expected value of the payload ID (V5) set on the local device is inconsistent with the transmission value on the opposite device.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the expected value of the payload ID (V5) set on the local device is inconsistent with the transmission value on the opposite device. If not, keep them consistent. Check whether the LP PLM alarm is cleared.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Chapter 4

System Alarms

Table of Contents

Cold Startup Notification.....	55
Active/Standby Card Switchover Notification	56
Backup Data Synchronization Failure Notification	56
CPU Overload Alarm	57
Memory Overload Alarm	57
Card Status Error Alarm	58
Terminal Login Notification.....	59
Terminal Logout Notification.....	59
NM Heartbeat Notification.....	59
MAC Table Utilization Exceeding Threshold Notification.....	60
Card Version Update Failure Alarm	61
Uplink Port Down Alarm.....	61
Access Port Up Notification.....	62
Access Port Down Notification	62
Uplink Port Output Bandwidth Occupancy Exceeding Threshold Alarm	63
Uplink Port Input Bandwidth Occupancy Exceeding Threshold Alarm	64
High Temperature Alarm.....	64
Fan Card Link Down Alarm.....	65
Fan Fault Alarm	65
Over-voltage Power Alarm	66
Under-voltage Power Alarm	66
NO Input Power Alarm.....	67
High Temperature Card Alarm	67
CCM Notification.....	68

Cold Startup Notification

Alarm Description When cold startup is implemented on ZXA10 C300, the system generates the notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
33025	Notification	Operation information

Impact on System During the equipment restart process, all the [ONU](#) services on the [PON](#) port are interrupted, the ONUs are disconnected. They recover automatically while the restart process is complete.

- Possible Causes** The main control card carries out the restart operation or the standby main control card is switched over to be the active main control card.
- Troubleshooting** Check whether any system configuration data is lost after the system is restarted.
- Result** The system is normal.

Active/Standby Card Switchover Notification

- Alarm Description** When the active/standby card switches over, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
33026	Notification	Operation information

- Effect on the System** None

- Possible Causes** The active/standby main control card switches over.

- Handling Processes** No need to handle.

- Result** The system is normal.

Backup Data Synchronization Failure Notification

- Alarm Description** When the system detects that backup data synchronization fails, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
33027	Notification	Fault alarm

- Impact on System** The active main control card fails to back up data to standby main control card in real time, but it has no effect on the current services. Part of the service data may be lost if the active/standby main control card switches over at this moment.

- Possible Causes** The fault may be caused by the following:
- The standby card fails to copy data from the active card.
 - The backplane is disconnected.

- Troubleshooting** To handle the fault, perform the following steps:

1. Reset the standby card.
2. Check the standby main control card. Make sure that main control card is in position and has a good contact with the back-plane.
3. Check whether the software versions of the active and standby cards are the same. If not, download the version file again.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The active/standby data is synchronized normally.

CPU Overload Alarm

Alarm Description When the CPU usage exceeds the threshold, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
33035	Warning	Fault alarm

Impact on System The device fails to run properly.

Possible Causes The CPU usage is too high or the CPU threshold is too low.

Troubleshooting To handle the fault, perform the following steps:

1. Observe the system service status. If numerous subscribers are using the services, it is not necessary to handle it.
2. Check whether the CPU overload threshold is too low. If yes, set the CPU overload threshold to a higher value.
3. Check whether the system is illegally attacked. Check the network security and ensure safe connection.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Memory Overload Alarm

Alarm Description When memory usage exceeds the threshold, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
33036	Warning	Fault alarm

Impact on System The device fails to run properly.

Possible Causes The memory usage is too high or the threshold is too low.

- Troubleshooting** To handle the fault, perform the following steps:
1. Observe the service status. If numerous subscribers are using the service, it is not necessary to handle it.
 2. Check whether the memory overload threshold is too low. If yes, set the memory overload threshold to a higher value.
 3. Check whether the system is illegally attacked. Check the network security and ensure safe connection.
 4. If the alarm still exists, contact the technical support personnel for assistance.
- Result** The alarm disappears.

Card Status Error Alarm

Alarm Description When the system detects that the card status is abnormal or cannot detect the card, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
33037	Critical	Fault alarm

Impact on System The card fails to work properly.

Possible Causes The fault may be caused by the following:

- The actual card type is inconsistent with the configured type.
- The card is pulled out or is offline.
- The inter-card communication is abnormal.
- The card hardware is faulty.

- Troubleshooting** To handle the fault, perform the following steps:
1. Log into the device in [NMS](#) or [CLI](#) mode to carry out the **reset card** command. Check whether the alarm is cleared.
 2. Check whether the heartbeat message is normal between the card and the main control card. If not, pull out the card and insert it again. Ensure that the card is in good contact with the backplane.
 3. Check whether the card type is correct.
 4. Check whether the main control card version is consistent with the line card version. If not, update the line card version.
 5. Replace the faulty card.
 6. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Terminal Login Notification

Alarm Description When the terminal device logs in to ZXA10 C300 through a console port or the out-of-band NMS, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
34816	Notification	Operation information

Impact on System None

Possible Causes It is caused by manual login.

Troubleshooting It is unnecessary to be handled.

Result The alarm disappears.

Terminal Logout Notification

Alarm Description When the terminal device logs out of ZXA10 C300 through a console port, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
34817	Notification	Operation information

Impact on System None

Possible Causes The fault may be caused by the following:

- It is caused by manual logout.
- The terminal is disconnected with ZXA10 C300.

Troubleshooting To handle the fault, perform the following steps:

- It is unnecessary to be handled if it is caused by Cause 1.
- If it is caused by Cause 2, check the connection between the terminal and ZXA10 C300. Make sure that the connection is normal.

Result The alarm disappears.

NM Heartbeat Notification

Alarm Description The NM heartbeat message is used to transmit the heartbeat message to the NM server periodically. Therefore, the NMS does not need to actively query the link condition between the NMS and the NE, which lowers the NMS load. This notification is not displayed on the NMS.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
34818	Notification	Operation information

Impact on System None**Possible Causes** The NE actively transmits the heartbeat message periodically to the NMS server.**Troubleshooting** Disable this alarm function.**Result** The alarm disappears.

MAC Table Utilization Exceeding Threshold Notification

Alarm Description When the usage of the MAC table on the main control card exceeds the threshold, the system generates this notification.**Alarm Property**

Alarm ID	Alarm Level	Alarm Type
35584	Notification	Fault alarm

Impact on System It may lead to service interruption.**Possible Causes** The fault may be caused by the following:

- The threshold is too low.
- The usage of the MAC table is too high.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the alarm threshold is properly configured. If not, modify the threshold value.
2. Delete some dynamic MAC addresses or shorten the aging time of the dynamic MAC addresses to accelerate the aging progress of the MAC addresses.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The usage of the MAC table is lower than the threshold.

Card Version Update Failure Alarm

Alarm Description When the administrator fails to update the card version, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
38401	Minor	Fault alarm

Impact on System The card version update fails.

Possible Causes The fault may be caused by the following:

- The NE does not support this card hardware type.
- The card type configured on the system does not match the card type reported by the line card.
- There is no card version file for update.
- The line card does not support the current line card version.
- Some errors occur in downloading.

Troubleshooting To handle the fault, perform the following steps:

1. Check the card. Make sure that the card and the main control card are in position and in good contact with the backplane and work properly. If not, carry out the update command again.
2. Update the registry editor for the NE card type.
3. Configure the card type again.
4. Download the corresponding version file to the main control card again. Carry out the update command again.
5. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Uplink Port Down Alarm

Alarm Description When the uplink port is disconnected, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
40713	Critical	Fault alarm

Impact on System The service on this uplink port is interrupted.

Possible Causes The physical line on the uplink port is faulty. The network cable or the optical fiber is improperly connected.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the uplink port cable is connected properly. If not, reconnect the cables or replace the cables.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Access Port Up Notification

Alarm Description When the link is connected on the ZXA10 C300 PON port, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
40714	Notification	Operation information

Impact on System None

Possible Causes Some ONUs are getting online on the PON port.

Troubleshooting No need to handle it.

Result The alarm disappears.

Access Port Down Notification

Alarm Description When the link is down on ZXA10 C300 PON port, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
40715	Notification	Fault alarm

Impact on System All the ONUs under this PON port fail to get online.

Possible Causes The fault may be caused by the following:

- This PON port is in **lock** or **shutdown** state.
- The optical fiber on this PON port is not well connected.
- The PON port has no bearing service or optical signal transmission.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the port is shut down. If yes, enable the PON port.
2. Check whether the optical fiber connected to this PON port is normal. If not, replace the optical fiber.

3. Check whether the system service configuration is normal. If not, configure the system data again.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Uplink Port Output Bandwidth Occupancy Exceeding Threshold Alarm

Alarm Description When the output bandwidth of the ZXA10 C300 uplink optical port exceeds the threshold, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
40718	Warning	Operation information

Impact on System It may lead to user service congestion and abnormal service.

Possible Causes The fault may be caused by the following:

- The threshold is incorrectly configured.
- Numerous users are using the services.
- The system is attacked by some illegal users.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the threshold is correctly configured. If not, modify the threshold.
2. Check the service usage condition in the system. If numerous users are using the services, it is unnecessary to handle the fault.
3. If the output bandwidth of the uplink port is continuously high, increase the upstream bandwidth.
4. Check whether the system is attacked by some illegal users. Ensure the network security and disconnect the illegal users.
5. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Uplink Port Input Bandwidth Occupancy Exceeding Threshold Alarm

Alarm Description When the input bandwidth of the ZXA10 C300 uplink optical port exceeds the threshold, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
40719	Warning	Operation information

Impact on System It may lead to user service congestion and abnormal service.

Possible Causes The fault may be caused by the following:

- The threshold is incorrectly configured.
- Numerous users are using the services.
- The system is attacked by some illegal users.

Troubleshooting

To handle the fault, perform the following steps:

1. Check whether the threshold is correctly configured. If not, modify the threshold.
2. Check the service usage condition in the system. If numerous users are using the services, it is unnecessary to handle the fault.
3. If the input bandwidth of the uplink port is continuously high, increase the upstream bandwidth.
4. Check whether the system is attacked by some illegal users. Ensure the network security and secure network connection.
5. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

High Temperature Alarm

Alarm Description When the system detects that the environment temperature exceeds the threshold, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
50944	Minor	Fault alarm

Impact on System The high environment temperature may damage the [NE](#) and thus affect the system services.

Possible Causes The fault may be caused by the following:

- The environment temperature is too high.

- Troubleshooting**
- The high temperature threshold is set improperly.
- To handle the fault, perform the following steps:
1. Check whether the high temperature threshold is set properly. If not, modify the threshold.
 2. Check the device installation environment and lower the temperature.
 3. Increase the fan speed.
 4. If the alarm still exists, contact the technical support personnel for assistance.
- Result** The alarm disappears.

Fan Card Link Down Alarm

Alarm Description When the system detects that the fan card link is down, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
50945	Minor	Fault alarm

Impact on System It may lead to NE temperature rising, and therefore damages the NE.

Possible Causes The fan card is not properly connected with the serial port line.

- Troubleshooting**
- To handle the fault, perform the following steps:
1. Check whether the fan card is properly connected with the serial port line and whether the serial line is normal.
 2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Fan Fault Alarm

Alarm Description When the system detects that the fan works improperly and fails to configure it properly, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
50948	Minor	Fault alarm

Impact on System The fan fault affects the system ventilation and leads to NE temperature rising, which may affect the system service and damage the NE.

Possible Causes The fan is damaged.

- Troubleshooting** To handle the fault, perform the following steps:
1. Replace the fan module.
 2. If the alarm still exists, contact the technical support personnel for assistance.
- Result** The alarm disappears.

Over-voltage Power Alarm

Alarm Description When the system detects that the input voltage of the equipment exceeds the over-voltage threshold, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
50952	Minor	Fault alarm

Impact on System Too high equipment input voltage may affect the system services and thus damage the [NE](#).

Possible Causes The power hardware is faulty.

- Troubleshooting** To handle the fault, perform the following steps:
1. Check whether the equipment input voltage is within the normal range.
 2. Check whether the equipment power card status is normal. If not, replace the power card.
 3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

Under-voltage Power Alarm

Alarm Description When the system detects that the input voltage of the equipment is lower than the set under-voltage threshold, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
50953	Minor	Fault alarm

Impact on System The system fails to work normally.

Possible Causes The power hardware is faulty.

- Troubleshooting** To handle the fault, perform the following steps:
1. Check whether the equipment input voltage is within the normal range.

2. Check whether the equipment power card status is normal. If not, replace the power card.
3. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

NO Input Power Alarm

Alarm Description When the system has no input power, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
50954	Minor	Fault alarm

Impact on System The system fails to work normally.

Possible Causes The power card is offline.

Troubleshooting To handle the fault, perform the following steps:

1. Insert the power card, and make sure that the power card is properly connected with the shelf.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

High Temperature Card Alarm

Alarm Description When the system detects that the card temperature exceeds the threshold, the system generates this alarm.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
50955	Minor	Fault alarm

Impact on System The high temperature of the card may affect the card services.

Possible Causes The fault may be caused by the following:

- The environment temperature is too high.
- The high temperature threshold is set improperly.
- The fan is damaged.

Troubleshooting To handle the fault, perform the following steps:

1. Check whether the high temperature threshold is set properly. If not, modify the threshold.

2. Check the device environment and lower the temperature.
3. Check whether the fan status is normal. If the fan is faulty, replace the fan card and increase the fan speed.
4. If the alarm still exists, contact the technical support personnel for assistance.

Result The alarm disappears.

CCM Notification

Alarm Description Each **MEP** set by the equipment transmit **CCMs** periodically. All the **MIPs** and **MEPs** in the maintenance domain can receive CCMs. When the system does not receive the CCM from a certain MEP in a period, the system generates this notification.

Alarm Property

Alarm ID	Alarm Level	Alarm Type
55552	Notification	Fault alarm

Impact on System The data link between the equipment and the MEP is unavailable. The services may be interrupted.

Possible Causes The data link is faulty.

Troubleshooting To handle the fault, perform the following steps:

1. Make sure that the physical connection is normal.
2. If the alarm still exists, contact the technical support personnel for assistance.

Result The CCM is normal.

Figures

Figure 1 Alarm Sound Setup	5
Figure 2 Alarm Color Setup	6
Figure 3 Forward Template Setup.....	7
Figure 4 New Severity	8
Figure 5 Modify Status	9
Figure 6 Maintenance Suggestion Setting.....	10
Figure 7 Function Buttons.....	10
Figure 8 Management Rule Setting.....	11
Figure 9 Perf Threshold Admin.....	12
Figure 10 Add Performance Threshold	12
Figure 11 Perf threshold admin.....	13
Figure 12 Real-Time Alarms	15
Figure 13 Real-Time Events	16
Figure 14 Real-Time Notifications.....	17
Figure 15 Current Alarm Query Conditions	18
Figure 16 Query Current Alarm.....	19
Figure 17 History Alarm Query Conditions.....	20
Figure 18 Query History Alarm	20
Figure 19 History Alarm Statistics — Report Management	21
Figure 20 History Alarm Statistics — List of Report Templates...	22
Figure 21 Query Event	23
Figure 22 Query Notification.....	24

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Tables

Table 1 Typographical Conventions	II
Table 2 Mouse Operation Conventions	III
Table 3 ZXA10 C300 Alarm List.....	2
Table 4 Button Description	10
Table 5 Parameter Description for Added Performance Threshold	13
Table 6 Fault Analysis For ONU Optical Module Diagnostic Parameter Alarm	32

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Index

A

Access Port Down Notification.....	62
Access Port Up Notification.....	62
Active/Standby Card Switchover Notification	56
Alarm Configuration	4
Alarm Information Query	17
Alarm Level	1
Alarm List.....	1
AU AIS	46
AU-LOP	46

B

Backup Data Synchronization Failure Notification	56
--	----

C

Card Status Error Alarm	58
Card Version Update Failure Alarm	61
CCM Notification.....	68
Cold Startup Notification.....	55
Collecting History Alarm Statistics.....	21
CPU Overload Alarm	57
Customizing Alarm Levels.....	7

D

Deny Unknown ONU Register	36
Device OLT PortBer State.....	25

E

Extended OAM	32
Extended OAM Discovery Failed	35

F

Fan Card Link Down Alarm	65
Fan Fault Alarm	65

H

High temperature Alarm	64
High Temperature Card Alarm	67
HP PLM	49
HP TIM.....	47
HP TU LOM	50
HP UNEQ	47
HP-AIS.....	49
HPRDI.....	48

L

Local ONU Configure Failed	32
Loss of Frame Alarm	42
LP PLM	54
LP TIM	52
LP UNEQ	52
LPRDI	53

M

MAC Table Utilization Exceeding Threshold Notification.....	60
Memory Overload Alarm	57
MS AIS	44
MS-RDI.....	44

N

NM Heartbeat Notification.....	59
NO Input Power Alarm.....	67

O

OLT Reset.....	31
ONU Dying Gasp	31
ONU Errored Frame Period Event	27-28
ONU Errored Frame Seconds Summary Event	28
ONU Errored Symbol Period Event.....	26
ONU Offline	30
ONU Optical Power Threshold Crossing	30
ONU Request Auto Config	37

ONU Uplink Bit Error	29	Unknown ONU Offline.....	36
ONU Uplink Frame Error	29	Unknown ONU Online.....	36
Optical Interface LOS		Uplink port down Alarm.....	61
Alarm	42	Uplink Port Input	
Out of Frame Alarm	43	Bandwidth Occupancy	
Over-voltage Power		Exceeding Threshold	
Alarm	66	Alarm	64
		Uplink Port Output	
		Bandwidth Occupancy	
		Exceeding Threshold	
		Alarm	63
P			
PDH Interface AIS Alarm.....	40		
PDH Interface LoS	39		
PON LOS	26		
PW Local Alarm	41		
PW Remote Alarm.....	41		
Q			
Querying Current Alarms.....	17		
Querying Events.....	22		
Querying History Alarms.....	19		
Querying Notifications	23		
R			
Real-Time Alarm			
Monitoring.....	14		
Real-Time Event			
Monitoring.....	15		
Real-Time Monitoring	14		
Real-Time Notification			
Monitoring.....	16		
RS SD.....	45		
RS SF	45		
S			
Setting Alarm			
Maintenance			
Suggestions	9		
Setting Alarm Preferences.....	5		
Setting Alarm Rules	11		
Setting the Alarm			
Threshold.....	12		
T			
Terminal Login			
Notification.....	59		
Terminal Logout			
Notification.....	59		
TU LOM.....	51		
TU-AIS.....	51		
TU-LOP	50		
U			
Under-voltage Power			
Alarm	66		

Glossary

AIS

- Alarm Indication Signal

AU

- Administrative Unit

AU-LOP

- Administrative Unit-Loss of Pointer

BER

- Bit Error Rate

CCM

- Continuity Check Message

CLI

- Command Line Interface

CPU

- Central Processing Unit

EPM

- Environment Power Monitoring

HP-AIS

- High Order Path Alarm Indication Signal

HPRDI

- High Order Path Remote Defect Indication

LOF

- Loss Of Frame

LOM

- Loss Of Multiframe

LOP

- Loss Of Pointer

LOS

- Loss Of Signal

LPRDI

- Low Order Path Remote Defect Indication

MAC

- Medium Access Control

MEP

- Maintenance association End Point

MIP

- Maintenance domain Intermediate Point

MS

- Multiplex Section

MS-AIS

- Multiplex Section - Alarm Indication Signal

MS-RDI

- Multiplex Section - Remote Defect Indication

- NE**
 - Network Element
- NM**
 - Network Management
- NMS**
 - Network Management System
- OAM**
 - Operation, Administration and Maintenance
- OLT**
 - Optical Line Terminal
- ONU**
 - Optical Network Unit
- PDH**
 - Plesiochronous Digital Hierarchy
- PLM**
 - PayLoad Mismatch
- PON**
 - Passive Optical Network
- PW**
 - Pseudo Wire
- RDI**
 - Remote Defect Indication
- RS**
 - Regenerator Section
- SD**
 - Signal Degrade
- SDH**
 - Synchronous Digital Hierarchy
- SF**
 - Signal Fail
- STM**
 - Synchronous Transport Module
- TDM**
 - Time Division Multiplexing
- TIM**
 - Trace Identifier Mismatch
- TU**
 - Tributary Unit
- TU-AIS**
 - Tributary Unit Alarm Indication Signal
- TU-LOP**
 - Tributary Unit-Loss of Pointer
- UNEQ**
 - UN-Equipped
- UNM**
 - Unified Network Management

VoIP

- Voice over Internet Protocol

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